

Position: Time-Series Foundation Models Require Explicit Domain-Level Benchmarks

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Abstract

Time series foundation models (TSFMs) have demonstrated strong performance on established benchmarks such as GIFT-Eval, Monash, and TSFM-Bench. However, these benchmarks pool datasets from many domains with uneven representation, which can obscure performance within specific application areas such as healthcare, finance, nature, retail, and transport. The necessity for domain-specific evaluation arises from the inherent structural diversity of time series data: clinical records often feature irregular sampling and informative missingness; financial sequences are characterized by high noise and stochastic trajectories; and environmental data, such as energy and weather, are governed by deterministic physical laws and strong seasonal hierarchies. Motivated by this heterogeneity, **we argue that TSFMs require explicit domain-specific benchmarks** so practitioners can reliably assess a model’s utility within their own application area. This is because cross-domain differences in data generation, sampling irregularity, and nonstationarity under concept drift fundamentally shape forecasting difficulty and failure modes. As a result, strong performance on aggregated leaderboards may not translate to reliable deployment within a specific domain. To test this, we evaluated seven TSFMs across 72 datasets from six domains and found substantial cross-domain variability. These findings confirm that global benchmark scores can be misleading and that domain-aware evaluations are essential for trustworthy TSFM selection.

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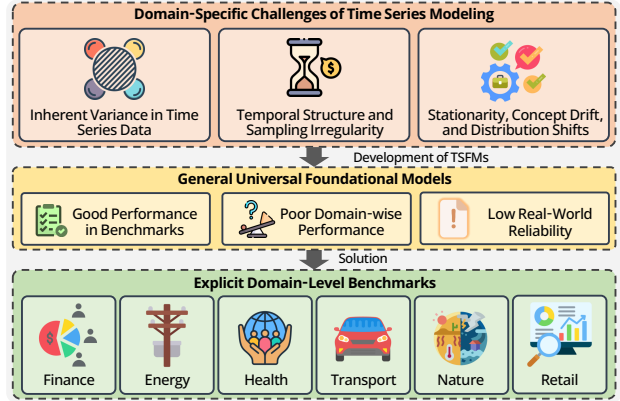


Figure 1. **Overview of our position.** Domain-specific differences in time series structure limit universal TSFMs, producing strong benchmark averages but inconsistent domain-wise reliability. We therefore argue for explicit domain-level benchmarks to expose transfer failures and enable meaningful evaluation.

1. Introduction

Inspired by breakthroughs in natural language processing and computer vision, where models like GPT (Brown et al., 2020), CLIP (Radford et al., 2021), and Gemini (Team et al., 2023) demonstrate remarkable zero-shot capabilities across diverse tasks, the machine learning community has begun pursuing **Time Series Foundation Models (TSFMs)** (Das et al., 2024b; Ansari et al., 2024) as a parallel paradigm. The central hypothesis is that a single, general-purpose model can be pretrained on large, diverse collections of time series to learn reusable temporal patterns. These patterns, such as seasonality, regime shifts, sparse events, and noise structures, appear consistently across different domains. To achieve this goal, TSFMs are trained on time series from finance, healthcare, mobility, and IoT sensors, aiming to provide the same benefits as language models: zero-shot forecasting, few-shot adaptation, and deployment without extensive feature engineering.

Several influential models have been developed following this vision, differing in their time series representations and scaling strategies while sharing a “pre-train once, use broadly” philosophy. TimesFM (Das et al., 2024b) adopts a GPT-style decoder-only Transformer architecture ($\approx 200\text{M}$ parameters) and is trained on large corpora (100B time points) for direct sequence modeling at scale.

Chronos (Ansari et al., 2024) takes a different approach by adapting language modeling techniques, discretizing continuous time series values into tokens to support probabilistic forecasting with models ranging from 20M to 710M parameters. Building on tokenization strategies, Moirai (Woo et al., 2024b) introduces patch-wise tokenization to improve efficiency and context coverage, training on billions of points. Time-MoE (Shi et al., 2025) pushes scaling further with 2.4B parameters via sparse expert routing. Other approaches include Lag-Llama (Rasul et al., 2023), which emphasizes probabilistic forecasting, and MOMENT (Goswami et al., 2024), which adapts a T5-like objective through masked time-series modeling. Despite these architectural and tokenization choices, all models converge on a shared principle: large-scale cross-domain pretraining acts as a universal temporal prior, allowing a single model to generalize across datasets and forecasting tasks with minimal per-application redesign.

However, time series data exhibit fundamentally different statistical and structural properties across domains, challenging the assumption that a single universal prior can transfer reliably. In healthcare, observations are often irregularly sampled and multi-rate. For example, high-frequency signals such as ECG may arrive at millisecond resolution, while laboratory measurements are recorded hourly or even daily, creating temporal structures that are difficult to reconcile with fixed-interval tokenization and patching (Li et al., 2025a; Rubanova et al., 2019; Horn et al., 2020). Moreover, clinical data frequently contain informative missingness, where the absence of a measurement itself carries diagnostic meaning, further complicating unified temporal encoding (Lipton et al., 2016). In finance, the dominant challenges are different: time series exhibit high volatility, near-random-walk behavior, and non-stationarity driven by regime shifts, market microstructure effects, and evolving participant behavior (Cont, 2001; Hamilton, 1989). High-frequency trading introduces event-driven sampling, violating uniform-interval assumptions altogether. More broadly, time series vary sharply in their generative mechanisms, predictability, temporal resolution, and signal-to-noise ratios. Energy patterns follow physical laws while retail demand reflects human behavior, atmospheric processes exhibit chaotic dynamics while utility loads remain stable, and temporal scales range from sub-second sensor measurements to monthly economic indicators. These differences indicate that time series domains often constitute distinct learning problems, not merely surface variations of a universal task (Gama et al., 2014; Lu et al., 2018).

To cope with this heterogeneity, TSFMs adopt architectural and preprocessing strategies to standardize inputs for a shared Transformer backbone. Patching divides series into fixed-length windows treated as tokens (Nie et al., 2023; Woo et al., 2024b), but assumes temporal continu-

ity and breaks down under irregular sampling. Channel-independence models each dimension separately, accommodating varying input variables but may discard cross-variable dependencies important in certain applications such as healthcare (Woo et al., 2024a). Instance normalization enables scale-invariant cross-domain transfer (Zeng et al., 2023), yet erases meaningful magnitude information such as absolute blood pressure thresholds unless reversible techniques are applied. Self-attention captures long-range dependencies, but its effectiveness depends on upstream tokenization choices. These design choices encode domain-specific inductive biases that fail outside their intended regimes. Patching acts as a low-pass filter unsuitable for high-frequency signals, while channel-independence may break down when cross-channel interactions are causal. Most critically, cross-domain pretraining induces gradient conflicts (Kouw & Loog, 2019): batches mixing medical series (requiring irregular timing sensitivity) with financial series (requiring regime-change robustness) produce conflicting updates that improve one domain while degrading another. The resulting representations reflect a compromise across incompatible objectives rather than a genuinely transferable temporal prior.

Motivated by this tension, **we argue that TSFMs require explicit domain-level benchmarks for meaningful assessment and deployment, since aggregated global leaderboards obscure domain-specific failure modes and produce misleading signals about real-world utility across diverse data-generating processes.** In this position paper, we explain why performance inconsistency emerges by linking it to domain-specific properties such as irregular sampling, non-stationarity, and heterogeneous noise that violate the assumptions embedded in current TSFM architectures and pretraining pipelines (§2). We also evaluate seven state-of-the-art TSFMs: TimesFM, Chronos, Moirai, Time-MoE, TimeGPT, Toto, and Sundial, across 72 datasets spanning six domains (healthcare, finance, energy, nature, transport, and retail), using consistent evaluation protocols for point forecasting accuracy (MSE, MAE) and ranking consistency (Demšar, 2006), showing that performance leadership remains fragmented rather than stable across domains (§4). Our findings call for domain-stratified benchmarks that surface application-specific strengths and weaknesses, enabling practitioners to select models with confidence and guiding the development of adaptive, hybrid TSFM designs that generalize reliably under real-world domain shift.

2. Domain-Specific Challenges of TSFMs

In this section, we explore the structural properties of time series data that create challenges for universal models, providing a foundation for understanding when domain-specific approaches may be preferable. We do not argue that univer-

sal models fail entirely, but rather that understanding these domain-specific utility is essential for practitioners deciding between universal and specialized approaches.

2.1. Inherent Variance in Time Series Data

Here, we explore the inherent variance in time series data that challenges the premise of a single universal temporal prior and motivates our position. Time series arising from physical, biological, and economic systems are governed by fundamentally different mechanisms, noise processes, and predictability horizons. For instance, an EEG voltage trace, an ICU vital-sign trajectory, and an equity return series emerge from distinct feedback loops and constraints. As a result, inductive biases that enable accurate forecasting in one domain often fail to transfer to others.

Presence of Deterministic Laws.

Many high-value forecasting domains are governed by deterministic laws with chaotic sensitivity. Large-scale atmospheric dynamics, for example, are described (under certain approximations) by the incompressible Navier–Stokes equations (Bistafa, 2024):

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} = -\frac{1}{\rho} \nabla p + \nu \nabla^2 \mathbf{u} + \mathbf{f}, \quad (1)$$

where $\mathbf{u}$ is velocity, p pressure, ρ density, ν viscosity, and $\mathbf{f}$ denotes external forcing. The shared physics across regions creates repeatable structure, enabling pre-training on weather or climate data to transfer smooth dynamical features across geographies and seasons. In contrast, financial markets are modeled as stochastic and partially adversarial environments, where exploitable patterns diminish through trading feedback. A canonical baseline is Geometric Brownian Motion (GBM), which models asset prices via the stochastic differential equation: $dS_t = \mu S_t dt + \sigma S_t dW_t$, where μ is the drift, σ the volatility, and W_t a Wiener process (Black & Scholes, 1973). Under strong forms of the Efficient Market Hypothesis (Fama, 1970), discounted prices behave as approximate martingales, making the current value a competitive forecast baseline and limiting persistent predictive structure. Because of this fundamental difference, aggregated benchmarks across domains can obscure domain-specific failures, making explicit domain-level evaluation essential for assessing transfer.

Statistical Markers. Several statistical markers further highlight how incompatible domains can be. The Hurst exponent H summarizes long-range dependence: $H > 0.5$ indicates persistent behavior, $H = 0.5$ corresponds to random-walk-like dynamics, and $H < 0.5$ indicates mean reversion. Geophysical and hydrological series often exhibit $H > 0.5$, reflecting long-memory variability, whereas liquid financial markets are frequently closer to $H \approx 0.5$, consistent with near-martingale behavior (Hurst, 1951; Lo, 1991). Signal-to-noise ratio (SNR) also separates regimes: physics-driven

domains often exhibit higher effective SNR, while high-frequency financial returns are dominated by noise, heavy tails, and regime shifts, making robust pattern extraction substantially harder (Mandelbrot & Hudson, 2007). Finally, positive Lyapunov exponents in chaotic systems bound predictability horizons, implying an intrinsic limit on how far ahead accurate forecasts are possible even with perfect models (Lorenz, 1963).

This variance has direct implications for TSFM pre-training. When a model is optimized over a heterogeneous corpus $\mathcal{D} = \bigcup_i \mathcal{D}_i$, gradients arising from different domains can conflict (Yu et al., 2020). In particular, domain-specific updates such as $\nabla_{\theta} \mathcal{L}_{\text{weather}}$ and $\nabla_{\theta} \mathcal{L}_{\text{finance}}$ may point in nearly orthogonal or opposing directions:

$$\langle \nabla_{\theta} \mathcal{L}_{\text{weather}}, \nabla_{\theta} \mathcal{L}_{\text{finance}} \rangle \approx 0 \quad \text{or} \quad < 0. \quad (2)$$

Such conflicts imply that improving performance in one generative regime can degrade another, encouraging convergence toward a statistical average that is suboptimal for both physics-driven and stochastic domains. This failure mode is structural, not merely a limitation of data scale or compute. It follows that claims of universal TSFM capability must be tested under explicit domain stratification, since aggregated benchmarks can obscure sharp domain-level inconsistencies. Domain-aware evaluation is therefore a prerequisite for understanding when universal pre-training helps, when it induces negative transfer, and when domain-conditioned or specialized foundations are required.

2.2. Temporal Structure and Sampling Irregularity

Time series are defined not only by observed values but also by the temporal process that generates observations. This temporal structure is strongly domain-dependent: domains encode meaning in both *what* is measured and *when* it is measured, so time assumptions that hold in one setting can fail in another, placing intrinsic limits on universal models.

Variance in Temporal Structure. Most forecasting pipelines and tokenization-based TSFMs implicitly assume a uniform time grid, $t_i = t_0 + ic$, equivalently enforcing constant spacing $\Delta t_i = t_{i+1} - t_i = c$ (e.g., 15 minutes or 1 hour). In practice, timestamps are often produced by an explicit *observation-time process*, and sampling gaps can be irregular, bursty, and history-dependent. A convenient abstraction models observation times as a point process: $t_1, t_2, \dots \sim \text{PP}(\lambda_{\mathcal{D}}(t) \mid \mathcal{H}_t)$, where $\mathcal{H}_t$ is the event history and $\lambda_{\mathcal{D}}$ captures domain-specific sampling mechanisms (Rubanova et al., 2019). The induced gaps, $\Delta t_i = t_{i+1} - t_i$, therefore, vary across domains and are frequently informative rather than random, especially in clinical settings (Lipton et al., 2016). This creates a temporal distribution shift: models trained under regular-time assumptions treat index distance as time distance, an inductive mismatch that

degrades cross-domain generalization.

Time Series Regularity Spectrum. We characterize domains along a *regularity spectrum* (Figure 2). Weather and energy often exhibit *homogeneous sampling* under standardized protocols (e.g., smart meters at 15-minute intervals), making fixed-length patch tokenization generally effective (Woo et al., 2024b; Rasul et al., 2023). In contrast, healthcare and high-frequency finance show *heterogeneous sampling* with highly variable, often event-driven observation patterns. In clinical data, ECG signals may be recorded at millisecond intervals while laboratory tests are conducted hours or days apart based on clinical judgment (Li et al., 2025a), a multi-rate structure that fundamentally challenges uniform-interval tokenization.

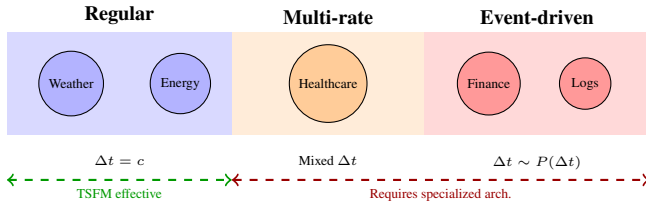


Figure 2. **Time Series Regularity Spectrum.** Sampling regimes vary across domains. Fixed patching works in regular settings, while multi-rate and event-driven data require time-aware or continuous-time biases.

This mismatch is particularly acute in ICU settings, where different physiological signals operate on vastly different timescales. ECG monitoring captures continuous waveforms at 100-500 Hz, while laboratory assays (e.g., blood chemistry panels) are ordered hours or days apart. These streams are rarely aligned and exhibit *informative missingness*, where the absence of a test reflects clinical judgment rather than random dropout (Lipton et al., 2016; Che et al., 2018). A TSFM trained primarily on regular sampling may treat such gaps as missing values to impute, potentially erasing semantically meaningful absence patterns. Fixed-length patching assumes uniform temporal intervals; when observation gaps vary from milliseconds to days, patch boundaries become arbitrary and fail to capture multi-rate temporal dynamics (Li et al., 2025a). This requires architectural redesign beyond standard patch-based approaches.

Spectral Content in Time Series Modeling. Domains also differ in *spectral content*. The Fourier transform,

$$X(f) = \int_{-\infty}^{\infty} x(t) e^{-2\pi i f t} dt, \quad (3)$$

highlights that macroeconomic indicators are dominated by low frequencies, while vibration and audio sensors contain high-frequency components. Deep networks exhibit *spectral bias*, learning low-frequency structure first and converging more slowly on high-frequency features (Rahaman et al., 2019); frequency-aware time series architectures have been proposed to mitigate this effect (Kang et al., 2024). As a result, TSFMs pretrained on low-frequency domains

can behave like *low-pass filters* when transferred to high-frequency sensors, smoothing the anomalies and transients they should detect. This spectral mismatch induces negative transfer that fine-tuning alone may not fully correct, since learned representations remain biased toward the frequency bands emphasized during pretraining.

2.3. Stationarity, Concept Drift, and Distribution Shifts

Stationarity. Generalization typically assumes that the training distribution P_{train} approximates the test distribution P_{test} . In time series, this assumption is routinely violated by *non-stationarity*, where statistical properties evolve over time. Critically, the form and severity of non-stationarity vary sharply across domains, so a single universal strategy for distribution shift is inherently misaligned with real-world forecasting and motivates our position that TSFMs require explicit domain-level benchmarks.

A series $\{X_t\}$ is *weakly stationary* if its mean $\mathbb{E}[X_t] = \mu$ and variance $\text{Var}(X_t) = \sigma^2$ are constant and its autocovariance $\gamma(h) = \text{Cov}(X_t, X_{t+h})$ depends only on lag h . Some domains partially approximate this behavior, such as energy loads with stable seasonal structure under a slowly varying envelope, but others violate it fundamentally. In finance, forecasting is often framed in terms of *log-returns*. Let $S(t)$ be the asset price and $X(t) = \ln S(t)$ its log-price; the return over horizon Δt is $r(t, \Delta t) = X(t + \Delta t) - X(t)$. Although $X(t)$ is frequently idealized as a random walk, empirical returns are far from i.i.d. Gaussian: linear autocorrelations are often weak, volatility is persistent (clustering), and returns are heavy-tailed (Cont, 2001). A TSFM pretrained on strongly periodic domains (e.g., energy) may therefore over-weight seasonality as a universal cue and misallocate capacity in finance, where time-varying volatility and shocks dominate.

Concept Drift and Distribution Shifts.

Beyond marginal changes, domains differ in whether the conditional relationship $P(Y|X)$ remains stable (Gama et al., 2014; Quiñero-Candela et al., 2009). Under *covariate shift*, $P(X)$ changes while $P(Y|X)$ is approximately invariant; for example, climate change may shift temperature distributions while pressure and wind physics persist. Under *concept drift*, $P(Y|X)$ itself evolves, as in finance, where indicator and return relationships change across regimes (Hamilton, 1989). The 2008 financial crisis illustrates this risk: long-standing cross-asset correlations inverted abruptly, rendering pre-crisis models not only inaccurate but overconfident.

Covariate shift and concept drift are both forms of dataset *distribution shift*, which occurs when training and deployment distributions diverge (Quiñero-Candela et al., 2009). These shifts vary systematically across domains. Finance

exhibits abrupt concept drift requiring frequent updates; energy shows predictable seasonal covariate shift suited to periodic recalibration; healthcare presents gradual patient-specific drift that challenges standard fine-tuning (Lu et al., 2019). A universal model cannot simultaneously trust stable physical relationships and remain skeptical of historically predictive patterns.

When TSFMs are pretrained on an aggregated corpus $\mathcal{D} = \bigcup_i \mathcal{D}_i$, they receive conflicting optimization signals, as gradients across domains are often nearly orthogonal. The resulting representation becomes a statistical average that lacks skepticism for rapid regime switches, adaptivity for patient-specific drift, and appropriate calibration when periodic domains dominate training. Fine-tuning cannot fully resolve this mismatch without catastrophic forgetting (Kouw & Loog, 2019), reinforcing that domain-aware evaluation is necessary for reliable deployment.

Takeaway. Together, these domain-specific challenges provide a theoretical basis for our central position that TSFMs do not yet behave as universal forecasters. The effective hypothesis class for forecasting varies across domains in causal structure, sampling process, spectral content, and drift dynamics, forcing a shared representation to absorb incompatible inductive biases. When pretraining aggregates these regimes, optimization becomes structurally conflicted, as gradients from physics-driven persistence, event-driven irregularity, and stochastic regime switching compete for the same parameters and produce statistically averaged representations rather than domain-optimal ones. These cross-domain failures are therefore not simply artifacts of limited scale or compute, but predictable outcomes of heterogeneous data-generating processes and distribution shift.

In the next section, we empirically validate our position by benchmarking leading TSFMs across six domains under a unified protocol, quantifying fragmentation in both accuracy and ranking stability.

3. Why Existing TSFM Benchmarks Fail?

Evaluating time series foundation models requires diverse benchmark suites that span multiple application domains. The three most prominent benchmarks such GIFT-Eval (Aksu et al., 2024), Monash (Godahewa et al., 2021d), and TSFM-Bench (Li et al., 2025b) collectively provide hundreds of datasets across domains including finance, healthcare, energy, transportation, and retail. However, closer examination reveals substantial domain imbalance in their composition.

Table 1 presents the domain distribution within GIFT-Eval and TSFM-Bench. Green highlights indicate the highest percentage within each column; red highlights indicate the lowest. GIFT-Eval, with 144,246 time series across

Table 1. Benchmark Data Statistics Aggregated by Domain.

Domain	GIFT-Eval		TSFM-Bench	
	# Series (%)	# Obs (%)	# Series (%)	# Obs (%)
Economic/Finance	100.0K (69.3%)	25.3M (16.0%)	236 (4.9%)	238.8K (0.6%)
Energy	2.0K (1.4%)	74.1M (46.9%)	493 (10.3%)	17.0M (42.2%)
Nature	32.6K (22.6%)	3.2M (2.0%)	54 (1.1%)	1.9M (4.8%)
Healthcare	1.0K (0.7%)	129.4K (0.1%)	955 (20.0%)	1.3M (3.3%)
Sales	3.7K (2.6%)	671.7K (0.4%)	—	—
Traffic/Transport	1.3K (0.9%)	38.0M (24.1%)	1.0K (21.6%)	18.2M (45.2%)
Web	3.5K (2.4%)	16.6M (10.5%)	2.0K (41.9%)	1.6M (3.9%)
Total	144.2K	158.0M	4.8K	40.2M

seven domains, is heavily skewed toward Economic/Finance (69.3% of series) and Nature (22.6%), leaving Healthcare with merely 0.7% representation and only 0.1% of total observations. TSFM-Bench exhibits different but equally problematic imbalance: Web data dominates at 41.9% of series, while Traffic/Transport contributes 45.2% of total observations. The benchmarks also show contrasting domain emphases: Economic /Finance represents just 0.6% of observations in the TSFM-Bench compared to 16.0% in GIFT-Eval, while Nature accounts for only 1.1% of the series in the TSFM-Bench versus 22.6% in GIFT-Eval. TSFM-Bench entirely omits Sales or Retail data. This inconsistent coverage means models evaluated on either benchmark alone may show inflated or deflated performance depending on their domain strengths. This domain imbalance motivates our stratified evaluation approach. Instead of aggregate scores, we group our 72 datasets into six balanced domains: Health, Finance, Nature, Transport, Retail, and Energy, so each contributes equally and enables fair cross-domain comparison across distinct data collection processes.

4. Empirical Validation

Experimental Design. We evaluate seven state-of-the-art TSFMs, namely TimesFM, Chronos, Moirai, Time-MoE, TimeGPT, Toto, and Sundial, across 72 datasets spanning six domains: healthcare, finance, energy, nature, transport, and retail. All models are benchmarked under a unified evaluation protocol, using point forecasting accuracy metrics (MSE and MAE) and ranking consistency following Demšar (2006). For fairness, each TSFM is evaluated in its strongest reported configuration, using the best-performing setting described in the original paper or official implementation. Additional details on the datasets, preprocessing, and implementation are provided in Appendix A.

4.1. Domain-wise Performance

Detailed domain-wise results are reported in Table 2 (energy) in the main paper, while Tables 5 (healthcare), 6 (finance), 7 (nature), 8 (transport), and 9 (retail) are provided in the appendix due to space constraints. On energy datasets (Table 2), TSFMs exhibit strong performance but still show fragmented leadership rather than a single consistently dominant model. Averaged across datasets

Table 2. Energy Time Series Forecasting Results

Dataset	Horizon	Chronos		Sundial		TimeGPT		TimesFM		TimesMOE		ToTo		Moirai	
		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
AUS Elec Demand	24	0.0965	0.2824	0.0452	0.1824	0.1628	0.3390	0.2049	0.4283	<u>0.0616</u>	<u>0.2238</u>	0.2572	0.4547	0.5435	0.6183
	48	0.2468	0.3903	0.0300	0.1471	0.2701	0.4754	0.1051	0.2829	<u>0.0403</u>	<u>0.1565</u>	0.4696	0.5836	0.3484	0.5248
	96	0.0899	0.2104	0.0233	0.1254	0.6010	0.6545	0.0725	0.2145	<u>0.0355</u>	<u>0.1301</u>	0.2735	0.4344	0.3047	0.4379
Electricity Weekly	24	1.8938	0.8021	4.0820	1.8735	1.6757	0.9854	1.9812	<u>0.7917</u>	<u>1.5199</u>	0.9913	0.5357	0.3301	1.8180	0.8297
	48	1.0033	0.4997	1.9888	1.1821	0.9494	0.7653	0.9929	<u>0.5020</u>	<u>0.8377</u>	0.7212	0.5051	0.3309	0.9518	0.5685
	96	1.5384	0.7182	1.3030	0.7318	1.4077	0.7351	1.5888	<u>0.7170</u>	<u>1.1375</u>	0.7835	0.4735	0.3294	1.4167	0.7237
ETTh1	24	0.2283	0.3865	0.2420	0.4044	<u>0.1807</u>	<u>0.3444</u>	0.3594	0.4581	0.1646	0.3250	3.0815	1.4634	0.4355	0.4844
	48	0.2586	0.4203	<u>0.2867</u>	<u>0.4248</u>	0.4537	0.4996	0.5055	0.5251	0.4087	0.4860	2.7232	1.3733	0.5730	0.5773
	96	0.4905	0.5491	0.2191	<u>0.3884</u>	0.3825	0.5045	0.2465	0.3595	0.2792	<u>0.4068</u>	2.4586	1.3231	0.3683	0.4273
ETTh2	24	0.3939	0.5651	0.4844	0.6033	0.2440	0.4401	<u>0.2169</u>	0.4138	0.1788	0.3762	0.3397	0.5180	0.2314	<u>0.4050</u>
	48	0.4504	0.5808	0.1561	0.3359	0.3284	0.4760	0.3249	0.4832	0.2606	0.4265	0.5246	0.6336	0.5369	0.6288
	96	0.3507	0.4583	0.6701	0.7525	0.2092	0.3461	0.2492	0.3887	<u>0.2126</u>	<u>0.3512</u>	0.7067	0.7373	0.4182	0.5028
ETTm1	24	0.4960	0.5354	0.3141	0.4137	0.5773	0.6387	0.2163	0.3580	<u>0.2348</u>	<u>0.3806</u>	0.6619	0.6351	0.7383	0.7413
	48	1.1027	0.8624	0.4640	<u>0.4702</u>	0.6325	0.6872	<u>0.3541</u>	0.4923	0.2976	0.4418	0.9816	0.8668	1.9918	1.2635
	96	2.3457	1.0366	0.2773	0.3990	1.2350	0.9271	0.1379	0.2574	<u>0.1871</u>	<u>0.3144</u>	0.8213	0.6837	2.9523	1.2133
ETTm2	24	0.0696	<u>0.2330</u>	0.2464	0.4649	0.0910	0.2769	0.1077	0.2769	0.1497	0.3213	<u>0.0811</u>	0.2314	0.1746	0.3590
	48	0.8794	0.8979	0.8366	0.7664	1.2731	1.0895	0.2987	0.4924	<u>0.4734</u>	<u>0.6398</u>	0.7530	0.8361	1.1056	0.9997
	96	0.3029	0.4796	0.2179	0.3873	0.3614	0.5333	<u>0.2192</u>	<u>0.3955</u>	0.2420	0.4093	0.2821	0.4785	0.3386	0.4840
London SmartMeters	24	<u>2.3292</u>	0.9403	2.8266	1.1401	3.5515	1.3026	2.7089	1.0717	1.9351	<u>0.9475</u>	5.8690	1.9358	2.6385	1.1145
	48	2.3505	0.9762	1.7465	0.8874	2.6237	1.0995	2.3544	0.9456	<u>2.0048</u>	<u>0.8904</u>	7.0594	2.0329	2.3622	0.9592
	96	<u>1.1737</u>	<u>0.6105</u>	1.2863	0.6569	1.5539	0.8628	1.2064	0.6105	1.0528	0.6025	6.7537	1.9636	1.3359	0.6795
Solar 10min	24	0.0001	0.0230	0.0034	0.0038	0.0087	0.0752	<u>0.0008</u>	0.0258	0.0041	0.0105	0.0087	0.0007	0.0069	<u>0.0027</u>
	48	0.1077	0.3185	<u>0.0035</u>	<u>0.0310</u>	0.1179	0.3414	0.0227	0.0966	0.0114	0.0791	0.0003	0.0103	0.1430	0.3469
	96	0.2472	0.4656	0.7006	0.5764	0.1308	0.3231	0.5050	0.4818	0.5495	0.5748	0.2578	0.4299	<u>0.2039</u>	<u>0.4181</u>
Average		0.7686	0.5518	0.7689	0.5562	0.7926	0.6134	0.6242	0.4612	0.5116	0.4579	1.4950	0.7757	0.9141	0.6379
1st Count		3	2	7	6	2	2	3	4	5	4	4	6	0	0

and horizons, TimesMoE achieves the best overall accuracy (MSE 0.5116, MAE 0.4579), outperforming TimesFM (MSE 0.6242, MAE 0.4612) and Chronos (MSE 0.7686, MAE 0.5518). However, wins are distributed across models: Sundial leads most often with 7 first-place finishes in MSE and 6 in MAE, while TimesMoE records 5 MSE wins and 4 MAE wins, indicating no universally stable leader even within a single domain. As additional experiments, the results for the two new models, TabPFN-TS and Moment (detailed in §B.2), are provided in the appendix and are not included in the overall ranking. For MOMENT, we evaluated it on the all domain, it performs well in Retail and Nature, matching top models on structured sales and climate data. TabPFN-TS demonstrates clear strengths in Health and Energy, securing the most first-place results in Health on clinical, hospital, and demographic datasets. These patterns extend across the remaining domains, reinforcing our position that TSFMs do not yet behave as universal forecasters and must be evaluated with explicit domain-level benchmarks.

4.2. Aggregated Performance and Ranking

Across domains, Figure 3 shows that TSFM rankings are highly non-uniform and rarely stable across application areas. No model stays consistently top-ranked across healthcare, finance, nature, transport, retail, and energy, and rank reversals are common. TimesFM performs strongly in retail but drops in finance and nature, while ToTo shows the opposite trend with comparatively better ranks in finance and

nature but weaker performance elsewhere. Even competitive models such as Chronos and TimesMOE exhibit wide spreads, indicating sensitivity to dataset choice and horizon within each domain. Overall, these distributions show that global averages can hide frequent domain-specific failures, supporting our position that aggregated leaderboards are unreliable indicators of deployment utility.

Figure 4 further confirms this fragmentation by revealing distinct domain performance footprints for each model. Rather than uniform strength, TSFMs show peaked behavior, with advantages concentrated in a few domains and clear weaknesses in others. TimesFM exhibits the most pronounced skew, with a strong retail advantage that does not transfer, while ToTo peaks in finance and nature but remains weaker in transport and retail. Sundial and Moirai show narrower gains, reinforcing that inductive biases tuned to particular temporal regimes do not generalize broadly. Together, these patterns affirm that TSFMs lack cross-domain consistency and motivate explicit domain-level benchmarks to evaluate transfer reliably.

4.3. Domain-Specific Performance Differences

Table 3 summarizes the Kruskal-Wallis test (Kruskal & Wallis, 1952) (detailed in §C) outcomes at horizon $h = 24$.

For both MAE and MSE, most models yield statistically significant evidence against the null hypothesis of domain-invariant performance, indicating that error distributions vary systematically across application domains rather than

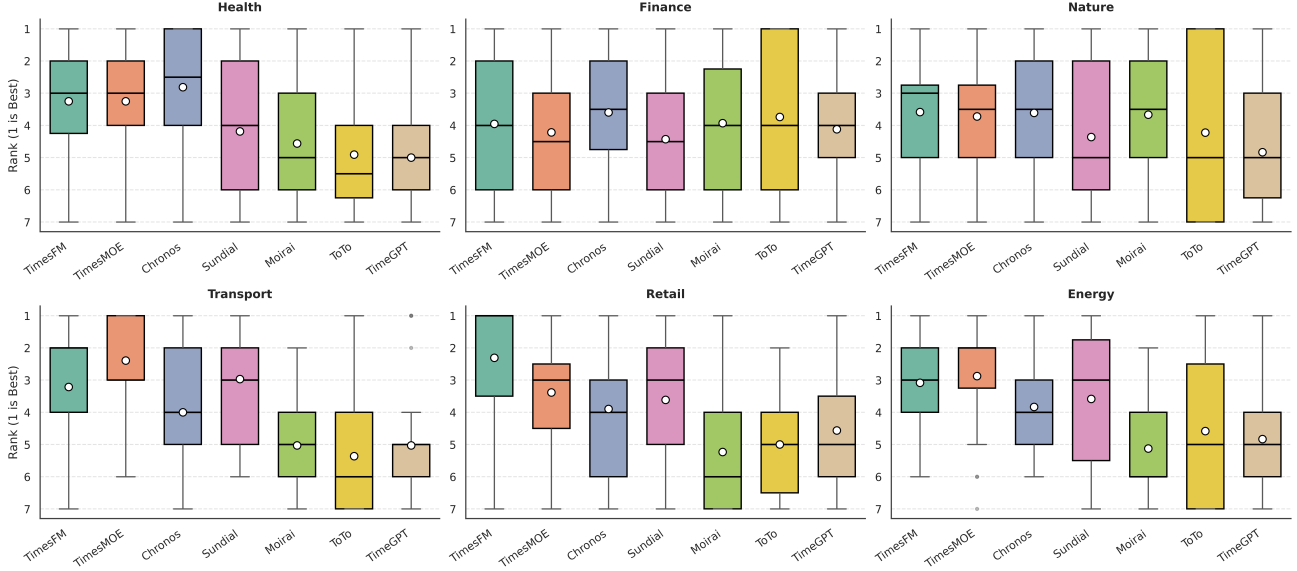


Figure 3. Model Rank Distribution (MAE) Across Domains. Box plots show the distribution of model rankings within each domain, where rank 1 indicates best performance. White circles denote mean rank; horizontal lines indicate median rank.

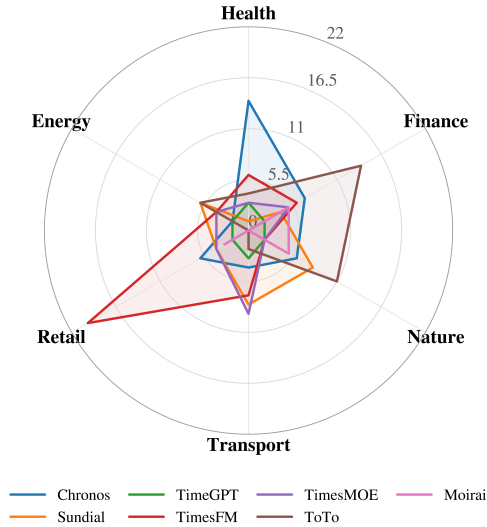


Figure 4. Domain-Wise Win Counts of Time-Series Foundation Models (MAE) (See Figure 6 for MSE rankings.)

remaining stable under a single global profile. The domain effect is generally stronger under MSE, where several models attain $p \leq 0.01$ (or better), suggesting that cross-domain shifts disproportionately amplify squared-error sensitivity through occasional high-magnitude deviations. MAE also shows consistent significance across models but at comparatively weaker levels, which aligns with its greater robustness to heavy-tailed errors and outlier behavior. Overall, these results provide quantitative evidence that time-series foundation model performance is *domain-dependent* at a fixed horizon, motivating domain-aware benchmarking and reporting beyond aggregate evaluations. More details are available in §C.2.

Table 3. Global domain effect at Horizon = 24 using the Kruskal-Wallis test. Reported values are p -values with significance markers: *ns* ($p > 0.05$), * ($p \leq 0.05$), ** ($p \leq 0.025$), *** ($p \leq 0.01$).

Model	MAE (p , sig.)	MSE (p , sig.)
Chronos (Ansari et al., 2024)	0.0444*	0.0146**
Moirai (Woo et al., 2024b)	0.0634 (<i>ns</i>)	0.0123**
Sundial (Liu et al., 2025)	0.0376*	0.0213**
TimeGPT (Garza et al., 2024)	0.0112**	0.0047***
TimesFM (Das et al., 2024a)	0.0190**	0.0110**
TimesMOE (Shi et al., 2025)	0.0118**	0.0048***
ToTo (Cohen et al., 2024)	0.0201**	0.0104**

5. Discussion

Our results provide empirical support for the theoretical argument developed in Section 2: **current TSFMs do not yet behave as universal forecasters**, because cross-domain transfer is constrained by structural differences in data generation, sampling, and drift. The observed performance fragmentation is systematic rather than noise around a shared optimum. Consistent rank reversals across domains indicate incompatible inductive biases: stable, seasonal domains (e.g., energy) favor models capturing smooth long-range structure, while stochastic, regime-shifting domains (e.g., finance) reward robustness to heavy tails, volatility, and rapidly changing $P(Y | X)$. A single shared representation must therefore encode conflicting assumptions about predictability and stationarity.

Importantly, this inconsistency persists even when TSFMs are evaluated in their strongest reported configurations. In the energy domain (Table 2), the best average model is TimesMoE (MSE 0.5116, MAE 0.4579), yet Sundial achieves the largest number of first-place finishes (7 in MSE

and 6 in MAE), demonstrating that leadership is unstable even within a comparatively structured domain. This instability becomes more pronounced when moving to cross-domain evaluation: Figure 3 shows wide rank distributions and frequent reversals across healthcare, finance, nature, transport, retail, and energy, indicating that global mean performance is an unreliable proxy for domain-level utility. The domain footprint in Figure 4 further reinforces that models specialize implicitly, with concentrated strengths in a subset of domains and clear weaknesses elsewhere, rather than expressing uniformly transferable capability. Finally, our Kruskal-Wallis analysis (Table 3) provides statistical evidence that error distributions differ systematically by domain for most models, rejecting the hypothesis of domain-invariant performance under a single global profile.

These findings reaffirm our position that **TSFMs require explicit domain-level benchmarks**. Aggregated leaderboards can overstate progress by averaging across domains where a model is well-aligned with the underlying temporal structure, while concealing failure modes in domains where its assumptions break. In practice, what matters is not whether a TSFM performs well on average, but whether it is reliable within the target application domain, under the relevant sampling regime, noise structure, and drift dynamics. Our evidence suggests that without domain-stratified evaluation, the community risks mistaking benchmark coverage for universality, and practitioners risk deploying models whose apparent global strength does not translate into consistent domain-level performance.

6. Alternative Views

Alternative View 1: Existing Benchmarks Already Provide Sufficient Granularity. One might argue that per-dataset reporting in current benchmarks already offers sufficient diagnostic resolution, making explicit domain-level evaluation unnecessary. However, Table 1 shows that domain representation is highly uneven, with some domains contributing far more datasets and observations than others. Consequently, aggregate metrics are dominated by well-represented domains, while systematic failures in sparsely represented but high-stakes domains remain hidden. Per-dataset results alone lack the structure needed to assess whether models generalize consistently across domains rather than excelling where data are abundant, conflating coverage with competence.

Alternative View 2: Universality Emerges Through Adaptation, Not Evaluation. A related view holds that expressive TSFMs should achieve cross-domain robustness via fine-tuning or in-context learning (ICL), obviating domain-level benchmarks. While adaptation can mitigate mismatch in stable, homogeneous domains, it fails under heterogeneous pretraining: cross-domain optimization yields sta-

tistical averages rather than transferable priors, and fine-tuning often induces catastrophic forgetting (Yang et al., 2024). Likewise, ICL cannot fix inductive misalignment when base representations are flawed (Dong et al., 2024). In fast-evolving domains such as finance (Ang & Timmermann, 2012) and health, requirements change faster than prompting or lightweight adaptation can accommodate (Das et al., 2025). Evidence from expert-combination frameworks further indicates that robust forecasting depends on specialization rather than universal representations (Ni et al., 2024).

Implication for Our Position. These alternatives raise valid expectations about benchmarks and adaptation, but neither resolves the core limitation. Without domain-stratified benchmarks, apparent gains may reflect dominance by over-represented domains rather than genuine transfer. Adaptation does not remove this need; it depends on prior diagnosis of where and why base models fail. Explicit domain-level evaluation is therefore a prerequisite for assessing universality claims and determining when adaptation suffices versus when domain-specific modeling is required.

7. Call to Action

Our theoretical analysis and empirical findings support a paradigm shift: the community should report *domain-stratified* benchmarking instead of only reporting pooled global evaluation. Current practice masks domain-level failure modes behind aggregate metrics, preventing diagnosis of where TSFMs succeed versus fail. We identify three priorities, ordered by immediacy of impact.

A1. Domain-Stratified Evaluation. It is the lowest-cost, highest-impact change. Existing benchmarks like Monash (Godahewa et al., 2021d), GIFT-Eval (Aksu et al., 2024), and TSFM-Bench (Li et al., 2025b) already span multiple domains; hence, the community also need to report results *by domain* rather than as pooled averages. Our analysis shows that models ranking first overall can rank last on specific domains; without stratified reporting, selecting an appropriate TSFM for a given application becomes difficult for practitioners.

A2. Evaluating Domain-Aware Models. Domain-specific benchmarks would enable fair comparison of emerging domain-aware models against universal TSFMs. Specialized models like MIRA (Li et al., 2025a) (targeting medical irregularity and missingness) and FinCast (Zhu et al., 2025) (addressing financial non-stationarity) report strong gains on their target domains but lack standardized evaluation against general-purpose alternatives. Stratified benchmarks would reveal whether domain-aware components consistently outperform global models within their intended domains, guiding future architectural investment.

A3. Cross-Domain Transfer Framework. TSFMs need a principled framework for cross-domain transfer, such as one that predicts which source domains benefit which targets. Currently, practitioners cannot know in advance whether a model pretrained mostly on energy data will help or harm medical forecasting. Such a framework would characterize domain similarity (e.g., by sampling frequency, stationarity, or causal structure) and identify when negative transfer is likely, enabling more informed model selection rather than repeated domain-specific trial-and-error tests (Wang et al., 2018; Kouw & Loog, 2019).

Software and Data

All datasets used in our experiments are publicly available benchmark datasets, and no proprietary or private data were used. Detailed descriptions of the datasets, preprocessing steps, and experimental setup are provided in the appendix. The time-series foundation models evaluated in this work are also publicly available through their official repositories or model releases. We use these public resources only for research and benchmarking purposes, following their respective licenses and usage terms.

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A. Extended Experimental Details

A.1. Dataset Description

We evaluate TSFMs on 72 datasets spanning six domains: Healthcare, Finance, Energy, Nature, Transport, and Retail. To ensure diversity and reliability, our datasets are sourced from established benchmarks and trusted repositories, including Monash Time Series Forecasting Archive (Godaheva et al., 2021d), GIFT-Eval (Aksu et al., 2024), TSFM-Bench (Li et al., 2025b), Kaggle competitions, the UCI Machine Learning Repository, and Zenodo. This multi-source approach maximizes coverage across temporal granularities, domain-specific patterns, and real-world forecasting challenges. Below we describe the datasets used in each domain.

A.1.1. HEALTHCARE

The healthcare domain comprises 11 datasets spanning diverse medical and epidemiological time series, including **COVID-19** (Li et al., 2025b) related intensive care admissions, mortality, and weekly case counts. **Hospital** records daily patient admission counts (Godaheva et al., 2021d). **Chickenpox** contains weekly chickenpox case counts from Hungary (Hun, 2021). **Illness** represents the percentage of patients with influenza-like illness (Godaheva et al., 2021d). **T1–T4** are heart rate time series from the MIT-BIH Arrhythmia Database (Moody & Mark, 2001). **US-Births** tracks daily birth records in the United States (Godaheva et al., 2021d).

A.1.2. FINANCE

The finance domain includes 14 datasets representing stock prices and financial indicators. Eleven individual stock time series (**AAL**, **AAPL**, **BLDP**, **CASH**, **CBD**, **DLR**, **EDUC**, **GWW**, **HAS**, **NSC**, **PHG**) are sourced from public financial data repositories (Li et al., 2025b). **CIF2016** originates from the Computational Intelligence in Forecasting competition (Štěpnička & Burda, 2015). **NN5-Daily** and **NN5-Weekly** contain ATM cash withdrawal data at daily and weekly granularities (Li et al., 2025b).

A.1.3. ENERGY

The energy domain consists of 8 datasets covering electricity consumption and generation patterns. **ETTh1**, **ETTh2**, **ETTm1**, and **ETTm2** provide hourly and 15-minute Electricity Transformer Temperature measurements (Li et al., 2025b). **AUS Elec Demand** tracks Australian electricity demand (Godaheva et al., 2021c). **Electricity Weekly** contains aggregated weekly electricity consumption (Godaheva et al., 2020a). **London SmartMeters** records household energy usage from UK smart meters (Godaheva et al., 2020b). **Solar 10min** captures solar power generation at 10-minute intervals (Godaheva et al., 2020d).

A.1.4. NATURE

The nature domain encompasses 12 datasets related to environmental and climate measurements. **Indian-Climate-AQI**, **Indian-Climate-Humidity**, **Indian-Climate-Temperature**, and **Indian-Climate-Wind** provide air quality and meteorological measurements from India.¹ **Jena-Climate-H2OC** and **Jena-Climate-RH** capture water vapor concentration and relative humidity from Jena, Germany (Wu et al., 2021). **Ozone**, **Pollution**, **Saugeen-Day**, **Sunspot**, **Temp-Hourly**, and **Weather** cover atmospheric ozone concentration, air pollution, daily river flow, solar activity, and hourly temperature and weather observations (Godaheva et al., 2021d).

A.1.5. TRANSPORT

The transport domain contains 11 datasets capturing traffic and mobility patterns. **Cross-1** through **Cross-6** track traffic at different road crossings (Axenie & Bortoli, 2020). **Pedestrian-Counts** records pedestrian traffic from sensors in Melbourne (Godaheva et al., 2020c). **Rideshare** captures rideshare demand data (Godaheva et al., 2021a). **Traffic-Hourly** and **Traffic-Weekly** provide San Francisco highway traffic volume at hourly and weekly temporal resolutions (Godaheva et al., 2020e;f). **Vehicle-Trips** records vehicle trip counts (Godaheva et al., 2021b).

¹<https://www.kaggle.com/datasets/ankushnarwade/indian-climate-dataset-20242025>

A.1.6. RETAIL

The retail domain includes 12 datasets representing product sales and demand. **Amazon Sales**² and **Coffee Sales**³ track e-commerce and retail transactions. **Online Retail** is a transactional dataset containing all transactions for a UK-based non-store online retailer (Chen, 2015). **Dominick’s Dataset** contains retail scanner data (Godahewa et al., 2021d). **Rossmann** contains daily sales from Rossmann drug stores.⁴ **Store 1 Item 1–Store 1 Item 5** and **Product 1–Product 3** capture item-level demand across retail locations.⁵

A.2. Implementation Details

A.2.1. EXPERIMENTAL SCOPE

All experiments are conducted under a **univariate forecasting** setting. Each time series is treated independently, without leveraging cross-series or multivariate dependencies. This choice ensures a fair comparison across TSFMs with different architectural assumptions regarding input dimensionality.

A.2.2. DATA PREPROCESSING

We apply a standardized preprocessing pipeline across all datasets. Each time series is reformatted to contain a `timestamp` column and a `value` column, ensuring compatibility across all evaluated TSFMs. Missing values are imputed using domain-appropriate methods: for finance, we apply forward-fill followed by backward-fill for leading gaps; for healthcare, we use forward-fill while preserving missingness indicators where clinically relevant; for energy and nature domains, we apply linear interpolation for short gaps and forward-fill for isolated missing points. We then apply instance-wise z-score normalization before model inference:

$$\hat{x}_t = \frac{x_t - \mu}{\sigma}, \quad \text{where} \quad \mu = \frac{1}{T} \sum_{t=1}^T x_t, \quad \sigma = \sqrt{\frac{1}{T} \sum_{t=1}^T (x_t - \mu)^2} \quad (4)$$

where x_t denotes the original value at time t , $\hat{x}_t$ is the normalized value, and T is the length of the time series. Predictions are denormalized using $x_t = \hat{x}_t \cdot \sigma + \mu$ before computing evaluation metrics. Finally, a chronological train/test split is used, with the test set comprising the final observations corresponding to the forecast horizon.

A.2.3. MODEL CONFIGURATIONS

We evaluate the publicly available variant of each TSFM to ensure maximum model capacity. All models are used in zero-shot mode with their default inference hyperparameters from the official implementations. Table 4 summarizes the model specifications and key inference parameters.

Table 4 reports the configurations of the time series foundation models evaluated in this study. For each model, we use the largest publicly available variant and perform inference in a zero-shot setting without task-specific fine-tuning. The table lists the total number of parameters, with activated parameters shown separately for mixture-of-experts models, enabling a direct comparison of model capacity. Maximum context length indicates the longest input sequence each model can process, which varies substantially across models and directly affects their ability to capture long-range temporal dependencies. All models are evaluated under identical forecasting horizons of 24, 48, and 96 steps to ensure comparability.

The inference parameters column summarizes the default hyperparameters provided by the official implementations, reflecting realistic out-of-the-box usage. Models such as TimesFM and Moirai support substantially longer contexts, while Chronos, Toto, and Time-MoE operate with more constrained input lengths. Time-MoE stands out with the largest total parameter count, although only a subset of parameters is activated during inference. TimeGPT is included as a proprietary baseline, for which architectural details and model size are not publicly disclosed. Overall, the table highlights the diversity in model scale, context capacity, and inference design among contemporary time series foundation models.

²<https://www.kaggle.com/datasets/rohiteng/amazon-sales-dataset>

³<https://www.kaggle.com/datasets/ihelon/coffee-sales>

⁴<https://www.kaggle.com/competitions/rossmann-store-sales/data>

⁵<https://www.kaggle.com/competitions/demand-forecasting-kernels-only/data>

Table 4. TSFM Model Configurations. All models use their largest available variant in zero-shot mode. *Params* denotes total parameters (activated parameters in parentheses for MoE models). *Context* denotes maximum supported input length. Inference parameters show default settings used in our evaluation. [†]Proprietary model with undisclosed architecture.

Model	Variant	Params	Max Context	Horizon	Inference Parameters
Chronos (Ansari et al., 2024)	2	710M	512	24, 48, 96	context_df, quantile_levels, future_df
TimesFM (Das et al., 2024b)	2.5	200M	16,384	24, 48, 96	max_context, normalize_inputs, infer_is_positive
Moirai (Woo et al., 2024b)	Large	311M	5000	24, 48, 96	prediction_length, windows, distance
Time-MoE (Shi et al., 2025)	Large	2.4B (1B)	4096	24, 48, 96	normed_seqs, max_new_tokens
Toto (Cohen et al., 2024)	Base	151M	512	24, 48, 96	prediction_length, num_samples, samples_per_batch
Sundial (Liu et al., 2025)	Base	128M	2880	24, 48, 96	seqs, max_new_tokens, num_samples
TimeGPT (Garza et al., 2024)	1	– [†]	– [†]	24, 48, 96	df, h, level

A.2.4. EVALUATION PROTOCOL

We evaluate all models at three forecast horizons: $h \in \{24, 48, 96\}$ time steps. Performance is measured using two standard metrics: Mean Squared Error (MSE) and Mean Absolute Error (MAE), computed on normalized predictions. All models are evaluated in a zero-shot setting without fine-tuning on the target datasets, reflecting practical deployment scenarios where labeled target data may be unavailable.

Mean Squared Error (MSE) is calculated as:

$$\text{MSE} = \frac{1}{T} \sum_{t=1}^T (y_t - \hat{y}_t)^2 \quad (5)$$

Mean Absolute Error (MAE) is calculated as:

$$\text{MAE} = \frac{1}{T} \sum_{t=1}^T |y_t - \hat{y}_t| \quad (6)$$

Here, T denotes the total number of time steps in the evaluation window, y_t represents the ground-truth value at time step t , and $\hat{y}_t$ denotes the corresponding model prediction. Both metrics are computed over normalized time series to ensure comparability across datasets and scales. MSE penalizes larger errors more strongly due to the squared term, while MAE provides a linear measure of average absolute deviation between predictions and true values.

B. Extended Experimental Results

B.1. Domain-wise Performance

Here we provide the detailed experimental results in Tables 5 (healthcare), 6 (finance), 7 (nature), 8 (transport), and 9 (retail).

Table 5. Health Time Series Forecasting Results

Dataset	Horizon	Chronos		Sundial		TimeGPT		TimesFM		TimesMOE		ToTo		Moirai	
		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
COVID-ICU	24	0.0001	0.0066	0.0005	0.0194	0.0004	0.0166	<u>0.0002</u>	<u>0.0077</u>	0.0034	0.0473	0.0003	0.0139	0.0003	0.0082
	48	0.0005	0.0218	0.0167	0.1117	0.0091	0.0877	<u>0.0031</u>	<u>0.0466</u>	0.0051	0.0636	0.0073	0.0845	0.0048	0.0557
	96	0.0010	0.0281	<u>0.0061</u>	<u>0.0570</u>	0.0457	0.1918	0.0128	0.0870	0.0136	0.0954	0.0295	0.1682	0.0498	0.1943
COVID-Deaths	24	<u>0.0041</u>	<u>0.0551</u>	0.0351	0.1626	0.0016	0.0353	0.0057	0.0710	0.0163	0.0964	0.0104	0.0783	0.0399	0.1884
	48	0.0547	0.2172	1.439	1.145	0.3511	0.5438	<u>0.0845</u>	0.2803	0.0884	<u>0.2720</u>	0.4304	0.6340	0.0539	0.1981
	96	0.8522	0.7216	3.167	1.476	2.305	1.251	<u>0.5924</u>	<u>0.6186</u>	0.3685	0.4696	4.000	1.996	1.935	1.1209
COVID-Weekly	24	0.0001	0.0054	0.0028	0.0410	0.0005	0.0181	<u>0.0002</u>	<u>0.0060</u>	0.0003	0.0144	0.0006	0.0225	0.0003	0.0140
	48	0.0002	0.0044	0.0009	0.0253	0.0011	0.0282	<u>0.0003</u>	<u>0.0149</u>	0.0005	0.0193	0.0014	0.0359	0.0011	0.0279
	96	0.0021	0.0406	0.0041	0.0463	0.0003	0.0142	0.0044	0.0556	0.0061	0.0706	0.0013	0.0349	<u>0.0006</u>	<u>0.0183</u>
Hospital	24	<u>0.4112</u>	0.5591	1.248	0.9643	0.4647	0.5541	0.3842	0.5134	0.4343	<u>0.5377</u>	0.5604	0.6274	2.0618	1.2402
	48	1.657	1.129	<u>1.020</u>	<u>0.7708</u>	2.747	1.537	2.813	1.553	0.6497	0.6268	2.024	1.278	1.2774	0.9515
	96	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Chickenpox	24	0.3785	0.2655	0.2881	<u>0.3217</u>	0.6204	0.6716	<u>0.3422</u>	0.4048	0.3517	0.3971	0.6278	0.4073	0.377	0.3856
	48	0.6658	0.4607	0.6885	0.5517	0.9805	0.6237	0.7382	0.5616	<u>0.6684</u>	<u>0.5068</u>	0.8786	0.8305	1.1987	0.765
	96	0.7367	0.5581	0.5438	<u>0.5352</u>	0.6447	0.6201	0.7808	0.6616	<u>0.5869</u>	0.5334	0.9126	0.8228	1.104	0.8446
Illness	24	0.8552	0.7422	<u>0.3515</u>	<u>0.4826</u>	4.535	1.839	0.2643	0.4435	0.5963	0.6646	1.193	1.065	0.0395	0.1738
	48	1.054	0.7755	0.8151	0.6681	2.927	1.153	<u>0.3505</u>	0.5079	0.3679	<u>0.4300</u>	0.1012	0.2754	0.7199	0.6396
	96	1.169	0.7321	1.504	0.8808	1.138	0.6773	1.440	0.7391	<u>0.6685</u>	<u>0.5303</u>	0.1003	0.2670	1.3324	0.7782
T1	24	2.010	1.330	<u>2.551</u>	<u>1.503</u>	3.017	1.618	3.610	1.796	2.966	1.618	2.820	1.619	1.6261	1.1483
	48	2.373	1.251	4.018	1.708	4.013	1.713	<u>2.860</u>	<u>1.389</u>	4.119	1.700	10.08	3.138	4.168	1.7356
	96	6.168	2.177	<u>4.395</u>	<u>1.857</u>	5.714	2.110	4.615	1.917	4.643	1.915	0.9504	0.9013	5.2957	2.0396
T2	24	0.2494	0.4765	<u>0.0461</u>	<u>0.1883</u>	0.0867	0.2363	0.0548	0.1935	0.1668	0.3823	0.0418	0.1845	0.0216	0.1201
	48	0.2431	0.4452	0.3220	0.4520	1.666	1.255	0.0817	0.2114	<u>0.1282</u>	<u>0.3124</u>	1.434	1.169	1.3475	1.0323
	96	1.305	0.8871	2.212	1.279	1.792	1.030	2.622	1.335	<u>1.107</u>	<u>0.7726</u>	0.1045	0.2885	2.8143	1.3836
T3	24	0.2049	0.3684	0.5265	0.6263	1.199	0.9873	<u>0.2653</u>	<u>0.4196</u>	0.3701	0.4903	0.3661	0.5157	0.3949	0.5175
	48	0.2911	0.4569	1.448	1.022	0.9223	0.8153	<u>0.4380</u>	0.5704	0.4590	<u>0.5617</u>	0.5474	0.6621	0.4111	0.4993
	96	0.4269	<u>0.5383</u>	0.6903	0.6692	1.363	1.028	<u>0.4290</u>	0.5325	0.5488	0.6403	0.4984	0.6158	0.7485	0.7591
T4	24	2.304	1.027	2.240	0.9859	1.529	0.7108	1.713	0.7820	<u>1.560</u>	<u>0.7140</u>	3.258	1.357	2.3415	1.0066
	48	0.8755	0.6161	1.069	<u>0.7500</u>	1.316	0.9674	<u>1.032</u>	0.7818	1.082	0.7764	1.595	0.7786	1.3398	0.9316
	96	0.8569	0.6936	1.140	0.8664	0.9973	0.8691	0.9698	0.8486	<u>0.9180</u>	<u>0.8165</u>	1.971	0.8792	1.1817	0.9565
US-Births	24	0.3915	0.4539	<u>0.3017</u>	<u>0.3802</u>	0.4426	0.4975	0.2088	0.3151	0.5670	0.5207	2.029	1.217	0.762	0.7179
	48	0.3528	0.3902	0.1916	0.2771	0.3767	0.3987	<u>0.2648</u>	<u>0.3081</u>	0.2949	0.3330	2.260	1.233	0.6377	0.6172
	96	0.6310	0.6572	0.2221	<u>0.2877</u>	0.7677	0.7373	0.1586	0.2209	<u>0.2148</u>	0.3073	3.726	1.459	1.6181	1.1015
1st Count		14	14	3	1	3	3	5	6	2	3	4	4	1	0

Position: Time-Series Foundation Models Require Explicit Domain-Level Benchmarks

Table 6. Finance Time Series Forecasting Results

Dataset	Horizon	Chronos		Sundial		TimeGPT		TimesFM		TimesMOE		ToTo		Moirai	
		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
AAL	24	0.0053	0.0657	0.0033	0.0483	<u>0.0008</u>	<u>0.0244</u>	0.0010	0.0275	0.0009	0.0250	0.0006	0.0203	0.0006	0.0203
	48	0.0025	0.0436	0.0040	0.0467	0.0048	0.0623	0.0008	0.0235	0.0235	0.1407	<u>0.0018</u>	<u>0.0333</u>	0.0073	0.0778
	96	0.0313	0.1532	<u>0.0023</u>	<u>0.0379</u>	0.0031	0.0413	0.0134	0.1015	0.0040	0.0518	0.0008	0.0229	0.0050	0.0564
AAPL	24	0.0375	0.1639	0.0578	0.2009	<u>0.0348</u>	<u>0.1559</u>	0.0567	0.2140	0.1929	0.4202	0.0599	0.2085	0.0190	0.1202
	48	<u>0.0236</u>	0.1344	0.0644	0.2080	0.0453	0.1767	0.0604	0.2162	0.0243	<u>0.1334</u>	0.0198	0.1024	0.0325	0.1569
	96	<u>0.0376</u>	<u>0.1569</u>	0.0675	0.2168	0.1425	0.3363	0.2342	0.4308	0.1998	0.4110	0.0352	0.1461	0.0698	0.2206
BLDP	24	0.0016	0.0326	0.0029	0.0503	<u>0.0009</u>	<u>0.0251</u>	0.0015	0.0341	0.0010	0.0268	0.0007	0.0218	0.0014	0.0332
	48	0.0086	0.0862	0.0311	0.1643	0.0038	0.0585	0.0028	0.0490	0.0112	0.1010	0.0013	0.0307	<u>0.0025</u>	<u>0.0447</u>
	96	0.0342	0.1516	0.0036	0.0533	0.0036	0.0541	<u>0.0022</u>	<u>0.0398</u>	0.0057	0.0692	0.0004	0.0148	0.0073	0.0733
CASH	24	0.0115	0.0897	0.0821	0.2188	0.0073	<u>0.0773</u>	0.0119	0.0899	0.0208	0.1322	<u>0.0084</u>	0.0711	0.0126	0.0984
	48	0.1731	0.3793	<u>0.1566</u>	<u>0.3547</u>	0.3984	0.5817	0.1608	0.3656	0.2659	0.4522	0.3532	0.5832	0.1207	0.3189
	96	0.2736	0.4385	0.3160	0.4802	0.2717	0.4352	0.5185	0.6876	0.2544	0.4231	0.0065	0.0703	<u>0.1731</u>	<u>0.3538</u>
CBD	24	0.0006	0.0209	0.0004	0.0173	0.0017	0.0350	<u>0.0005</u>	<u>0.0180</u>	0.0022	0.0404	0.0008	0.0224	0.0025	0.0452
	48	0.0004	0.0157	0.0018	0.0338	0.0013	0.0289	0.0040	0.0586	0.0030	0.0448	0.0014	0.0316	<u>0.0010</u>	<u>0.0242</u>
	96	0.0017	0.0312	0.0046	0.0591	0.0053	0.0665	<u>0.0032</u>	<u>0.0477</u>	0.0037	0.0524	0.0035	0.0560	0.0049	0.0630
CIF2016	24	0.2809	0.4653	0.3568	0.5394	0.3391	0.5058	0.1089	0.2864	0.3060	0.4945	<u>0.1586</u>	<u>0.3561</u>	0.3215	0.4214
	48	0.3526	0.5027	1.2910	0.9559	1.0906	0.9289	0.3418	0.4569	0.1832	0.3402	<u>0.2344</u>	<u>0.4281</u>	0.8567	0.9413
	96	<u>1.8719</u>	<u>1.1524</u>	3.7827	1.7269	2.7886	1.4656	2.4008	1.2885	0.7312	0.6748	5.6230	2.3614	1.4563	1.2345
DLR	24	<u>0.0488</u>	<u>0.2035</u>	0.0491	0.2073	0.0826	0.2738	0.0629	0.2371	0.0895	0.2868	0.0826	0.2828	0.0330	0.1649
	48	0.0280	0.1449	0.0317	0.1609	0.0284	0.1386	0.0242	0.1255	0.0254	0.1311	0.0089	0.0780	<u>0.0201</u>	<u>0.1130</u>
	96	0.2897	0.4550	<u>0.1533</u>	<u>0.3107</u>	0.3008	0.4658	0.4132	0.5442	0.1310	0.2941	0.1565	0.3892	0.3592	0.5227
EDUC	24	<u>0.0171</u>	<u>0.1053</u>	0.0057	0.0637	0.0340	0.1602	0.0229	0.1251	0.0360	0.1592	0.0337	0.1694	0.0277	0.1394
	48	0.0091	0.0790	0.0143	0.0964	0.0207	0.1107	<u>0.0138</u>	<u>0.0909</u>	0.0359	0.1464	0.0644	0.2364	0.0150	0.0942
	96	0.0664	0.2095	<u>0.0305</u>	<u>0.1372</u>	0.0584	0.2008	0.2466	0.4303	0.0597	0.2066	0.0105	0.0878	0.0553	0.1890
GWW	24	0.0128	0.0945	0.0097	0.0811	<u>0.0076</u>	<u>0.0718</u>	0.0054	0.0615	0.3383	0.5193	0.0089	0.0824	0.0083	0.0751
	48	0.0728	0.2503	<u>0.1484</u>	<u>0.3281</u>	0.1917	0.3721	0.2655	0.4385	0.5015	0.6005	0.2989	0.5372	0.3260	0.4878
	96	0.4659	<u>0.6082</u>	0.8045	0.8327	0.5076	0.6442	<u>0.4650</u>	0.6119	1.3606	1.0952	0.8717	0.9304	0.4011	0.5502
HAS	24	0.0100	0.0760	0.0271	0.1508	0.0129	0.0922	0.0160	0.1094	0.0082	0.0699	0.0064	0.0658	<u>0.0066</u>	<u>0.0671</u>
	48	0.1381	0.3264	0.0118	0.0886	0.1469	0.3490	0.1382	0.3356	0.0927	0.2849	0.1300	0.3552	<u>0.0660</u>	<u>0.2184</u>
	96	0.1794	0.3374	0.3867	0.5204	<u>0.1808</u>	<u>0.3407</u>	0.2645	0.4120	0.2886	0.4294	0.4007	0.6301	0.2458	0.3948
NN5-Daily	24	0.2038	0.3978	0.4372	0.5632	1.3864	1.0229	<u>0.3319</u>	<u>0.5045</u>	0.4215	0.5639	1.3616	0.8480	1.3150	0.9894
	48	0.2289	0.3955	0.4743	0.5437	1.2907	0.9994	<u>0.2553</u>	<u>0.4253</u>	0.3801	0.5077	1.3661	0.8855	1.3816	1.0216
	96	<u>0.3368</u>	<u>0.4489</u>	0.6484	0.6649	1.2380	0.9543	0.2910	0.4174	0.4015	0.4756	1.4396	0.8693	1.2418	0.9454
NN5-Weekly	24	0.9215	0.8195	1.0706	0.8935	0.1581	0.2651	1.3840	0.9978	<u>0.8641</u>	<u>0.7884</u>	0.8650	0.8788	0.8791	0.9854
	48	2.0686	1.3107	3.1269	1.6339	<u>1.4597</u>	<u>1.1442</u>	1.7961	1.2018	1.3692	1.0402	3.7536	1.8882	1.9675	1.3456
	96	1.4173	0.9335	2.5851	1.3653	0.3537	0.5269	2.1923	1.2239	<u>0.9116</u>	<u>0.7517</u>	3.1000	1.7027	1.2535	0.9453
NSC	24	<u>0.0282</u>	<u>0.1542</u>	0.1690	0.3806	0.0611	0.2327	0.0153	0.1086	0.0710	0.2509	0.0401	0.1878	0.1516	0.3722
	48	0.1163	0.2807	0.1000	0.2571	<u>0.0561</u>	<u>0.1989</u>	0.0240	0.1311	0.0793	0.2408	0.1260	0.3442	0.0779	0.2444
	96	0.0877	0.2270	0.0684	<u>0.2067</u>	<u>0.0575</u>	0.2087	0.1397	0.3038	0.1015	0.2676	0.0165	0.0989	0.0805	0.2302
PHG	24	0.0262	0.1520	0.0075	0.0745	0.0158	0.1174	<u>0.0114</u>	<u>0.0987</u>	0.0169	0.1212	0.0224	0.1442	0.0146	0.1158
	48	0.0648	0.2455	0.0504	0.2160	0.0558	0.2228	0.0529	0.2197	<u>0.0381</u>	<u>0.1824</u>	0.0247	0.1514	0.0438	0.1968
	96	0.2699	0.4631	0.4384	0.6096	0.1902	<u>0.3950</u>	0.3481	0.5234	0.0510	0.1982	<u>0.1769</u>	0.4141	0.3806	0.5656
1st Count		7	7	4	4	3	2	6	6	5	5	13	14	5	5

Table 7. Nature Time Series Forecasting Results

Dataset	Horizon	Chronos		Sundial		TimeGPT		TimesFM		TimesMOE		ToTo		Moirai	
		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
Indian-Climate-AQI	24	1.2535	1.0087	1.4445	1.0758	1.1318	0.9752	1.0533	0.9345	<u>1.0427</u>	<u>0.9250</u>	1.4818	1.1322	0.9997	0.8982
	48	1.2384	0.9527	1.4944	1.0679	0.9956	<u>0.8847</u>	1.0496	0.9232	1.0434	0.9221	1.3549	1.1162	<u>1.0307</u>	0.8783
	96	<u>0.9693</u>	0.8384	1.0595	0.8772	0.9613	0.8661	0.9859	0.8692	1.0035	0.8703	1.3557	1.0966	1.0041	<u>0.8614</u>
Indian-Climate-Humidity	24	0.9298	0.8049	1.0453	0.8721	1.1608	0.9118	<u>0.9672</u>	0.8456	0.9780	0.8506	1.0461	<u>0.8285</u>	1.5114	1.0255
	48	1.9417	1.1605	1.3214	0.9953	1.1658	0.9206	1.1307	0.9075	<u>1.0597</u>	<u>0.8951</u>	0.7978	0.7421	1.1569	0.9423
	96	1.1853	0.9044	1.3474	1.0075	1.4194	1.0328	1.0577	<u>0.8785</u>	<u>1.0365</u>	0.8834	0.8519	0.7904	1.1362	0.9141
Indian-Climate-Temperature	24	1.5446	1.0352	1.1595	<u>0.8897</u>	1.1297	0.9557	1.1244	0.9628	0.9712	0.8907	1.3352	1.0130	<u>0.9936</u>	0.8795
	48	0.9568	0.8437	1.2422	0.9089	1.2551	0.9841	1.0163	0.8704	<u>0.9577</u>	<u>0.8486</u>	1.0778	0.9446	1.1050	0.9242
	96	1.0189	<u>0.8541</u>	1.2117	0.9059	0.9511	0.8391	<u>0.9886</u>	0.8589	0.9921	0.8624	1.1640	0.9788	1.0203	0.8693
Indian-Climate-Wind	24	1.1475	0.9408	1.6223	1.0837	<u>1.2268</u>	<u>0.9795</u>	1.4137	1.0355	1.3093	1.0061	1.3958	1.0166	1.3573	1.0713
	48	1.2893	0.9736	<u>1.0953</u>	0.8603	1.0591	<u>0.8829</u>	1.3218	0.9751	1.1367	0.9160	1.3307	1.0291	1.1492	0.9080
	96	1.1354	0.9305	1.2724	0.9611	1.0642	<u>0.8905</u>	1.1910	0.9404	<u>1.0562</u>	0.9007	1.3350	0.9999	0.9986	0.8624
Jena-Climate-H2OC	24	0.2499	0.4587	0.4141	0.6091	0.8181	0.7452	0.1196	0.2941	0.6825	0.7904	0.3448	0.5249	<u>0.1396</u>	<u>0.3141</u>
	48	<u>0.0553</u>	0.1927	0.1163	0.2920	0.4884	0.5097	0.0534	<u>0.1971</u>	0.3833	0.5685	0.1595	0.3433	0.0945	0.2342
	96	0.9002	0.8096	1.4743	1.1173	0.5016	<u>0.6170</u>	0.7415	0.7369	0.7109	0.6970	0.2429	0.4599	<u>0.3653</u>	0.7543
Jena-Climate-RH	24	0.0048	0.0488	0.0109	0.0959	0.0727	0.2453	<u>0.0025</u>	<u>0.0467</u>	0.0015	0.0344	0.0093	0.0838	0.0484	0.1599
	48	0.0404	0.1739	0.0060	0.0663	0.1896	0.3751	<u>0.0115</u>	<u>0.0868</u>	0.0227	0.1185	0.0269	0.1583	0.0271	0.1466
	96	0.0238	0.1279	0.3168	0.4568	0.2204	0.3383	0.0399	0.1551	<u>0.0195</u>	<u>0.1156</u>	0.0108	0.0842	0.0454	0.1870
Ozone	24	0.2684	0.3869	0.4698	0.5519	0.4124	0.5698	0.2084	0.3566	<u>0.2106</u>	<u>0.3642</u>	0.4294	0.5880	0.2930	0.4461
	48	0.4138	0.5323	0.6862	0.7233	0.4398	0.6672	0.8674	0.8274	<u>0.3964</u>	0.5426	0.1825	0.3801	0.4041	<u>0.5233</u>
	96	0.3072	<u>0.4635</u>	0.3769	0.4853	0.4428	0.6898	0.7489	0.7184	0.3871	0.5044	<u>0.3216</u>	0.4539	1.4033	1.0576
Pollution	24	<u>0.0080</u>	<u>0.0831</u>	0.0069	0.0632	0.4267	0.6149	0.0138	0.1056	0.1909	0.3834	0.0327	0.1496	0.0140	0.1041
	48	1.7165	1.2588	0.0802	0.2120	1.9556	1.3346	<u>0.1772</u>	<u>0.3087</u>	0.9489	0.8874	1.2693	1.0830	1.3063	1.0443
	96	4.8114	2.0212	0.3613	0.4700	2.0119	1.3119	0.8198	0.6366	1.0054	0.8369	<u>0.4076</u>	<u>0.6107</u>	7.0588	2.4425
Saugeen-Day	24	1.3769	0.6235	1.6836	0.7641	1.5672	0.7099	1.8297	0.7851	<u>1.5489</u>	<u>0.6811</u>	2.4603	0.9531	1.6676	0.7462
	48	1.4326	0.7790	1.2216	0.6715	1.1070	0.6083	1.2939	0.7005	<u>1.1359</u>	<u>0.6287</u>	2.7519	1.0690	1.2425	0.6706
	96	0.8872	0.6055	<u>0.6768</u>	<u>0.4474</u>	1.0457	0.7056	0.8490	0.5861	0.6211	0.4027	3.2163	1.2856	1.0721	0.7186
Sunspot	24	0.0010	<u>0.0065</u>	0.0003	0.0168	0.0412	0.1827	<u>0.0002</u>	0.0096	0.0024	0.0416	0.0001	0.0020	0.0017	0.0216
	48	0.0078	0.0326	<u>0.0071</u>	0.0677	0.1716	0.3862	0.0075	0.0330	0.0074	0.0604	0.0032	0.0021	0.0079	<u>0.0309</u>
	96	<u>0.0052</u>	0.0338	0.0136	0.1076	0.3291	0.4921	0.0059	0.0310	0.0075	0.0719	0.0043	0.0121	0.0061	<u>0.0285</u>
Temp-Hourly	24	0.1469	0.3544	0.0246	0.1328	0.0772	0.2397	<u>0.0552</u>	<u>0.2158</u>	0.0792	0.2585	1.5738	1.1569	0.1346	0.3246
	48	<u>0.1511</u>	<u>0.3281</u>	0.0988	0.2626	0.2287	0.4173	0.1969	0.3916	0.1753	0.3558	1.3998	1.1113	0.1582	0.3421
	96	0.2937	0.4776	0.2167	0.3965	0.4062	0.5706	0.3809	0.5494	<u>0.2469</u>	0.4369	1.5869	1.1484	0.2811	<u>0.4366</u>
Weather	24	0.0999	0.1352	0.1752	0.2668	0.0987	0.1974	0.1004	<u>0.1321</u>	<u>0.0889</u>	0.2382	0.0367	0.0850	0.1005	0.1334
	48	0.0576	0.0878	0.1767	0.2184	0.1467	0.2132	0.0568	0.0856	0.0684	0.2279	0.0370	<u>0.0850</u>	<u>0.0564</u>	0.0839
	96	0.0813	0.1028	0.1916	0.2851	0.1532	0.2791	<u>0.0803</u>	0.1065	0.0805	0.2176	0.0417	0.0843	0.0819	<u>0.1008</u>
1st Count		5	6	7	8	5	2	3	2	3	2	11	11	2	5

Table 8. Transport Time Series Forecasting Results

Dataset	Horizon	Chronos		Sundial		TimeGPT		TimesFM		TimesMOE		ToTo		Moirai	
		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
Cross-1	24	0.3452	0.5044	0.0215	0.1197	0.4157	0.5736	<u>0.0220</u>	<u>0.1267</u>	0.0268	0.1404	0.0712	0.2108	0.4765	0.6049
	48	0.3358	0.4755	0.0236	0.1173	0.4693	0.5352	<u>0.0257</u>	<u>0.1276</u>	0.0294	0.1384	0.3931	0.5908	1.2434	0.8874
	96	0.3069	0.4570	<u>0.0520</u>	<u>0.1833</u>	0.6788	0.7031	0.0500	0.1693	0.0604	0.2015	0.9757	0.9730	0.6132	0.7061
Cross-2	24	0.3981	0.5479	<u>0.0617</u>	<u>0.1967</u>	0.6959	0.7674	0.0533	0.1833	0.0766	0.2309	0.1107	0.2909	0.6646	0.7308
	48	1.3764	0.9720	0.0495	0.1727	0.6755	0.6730	<u>0.0537</u>	<u>0.1734</u>	0.0635	0.1901	0.2915	0.5042	0.8898	0.8044
	96	0.9861	0.8128	0.1497	0.3121	0.9454	0.8112	<u>0.1091</u>	0.2433	0.0992	<u>0.2484</u>	1.5881	1.2486	0.2623	0.3568
Cross-3	24	0.5020	0.6194	0.0286	0.1488	0.4326	0.5809	<u>0.0297</u>	<u>0.1489</u>	0.0404	0.1664	0.1545	0.2994	0.4579	0.6097
	48	0.6596	0.6678	0.0477	0.1798	0.3139	0.4548	<u>0.0511</u>	<u>0.1812</u>	0.0760	0.2237	0.2094	0.3547	0.3252	0.4715
	96	0.1163	0.2750	0.0770	0.2314	0.2835	0.4346	0.4512	0.5734	<u>0.0992</u>	<u>0.2538</u>	0.4514	0.5870	0.1609	0.3177
Cross-4	24	0.4440	0.5210	<u>0.0693</u>	0.2045	0.4153	0.5287	0.0610	0.1900	0.0696	<u>0.2014</u>	0.1313	0.3197	0.3815	0.5288
	48	0.5489	0.6258	0.0959	<u>0.2378</u>	0.4645	0.5961	0.0873	0.2351	<u>0.0939</u>	0.2396	0.3080	0.4929	0.5453	0.6310
	96	0.4706	0.5483	0.1075	0.2554	0.5529	0.6049	<u>0.1099</u>	<u>0.2613</u>	0.1113	0.2684	0.9300	0.9153	0.3004	0.4227
Cross-5	24	0.3848	0.5093	<u>0.0660</u>	<u>0.1943</u>	0.3238	0.4799	0.0602	0.1825	0.0686	0.1946	0.0982	0.2415	0.2686	0.4445
	48	0.2994	0.4534	0.0790	0.2221	0.3013	0.4558	0.1091	0.2561	<u>0.0865</u>	<u>0.2245</u>	0.1617	0.3371	0.3707	0.5045
	96	0.6295	0.6137	<u>0.1332</u>	<u>0.2692</u>	0.7740	0.6939	1.3262	1.0007	0.0818	0.2251	0.2641	0.4402	0.5839	0.5806
Cross-6	24	0.2387	0.3872	<u>0.0543</u>	<u>0.1774</u>	0.2949	0.4756	0.0572	0.1802	0.0480	0.1622	0.1161	0.2775	0.3042	0.4700
	48	0.5090	0.6210	<u>0.0519</u>	<u>0.1841</u>	0.3152	0.4816	0.0996	0.2760	0.0449	0.1655	0.2729	0.4750	0.3284	0.4774
	96	0.1163	0.2750	<u>0.0770</u>	<u>0.2314</u>	0.2835	0.4346	0.4512	0.5734	0.0755	0.2147	0.7645	0.8331	0.1254	0.2839
Pedestrian-Counts	24	0.0014	<u>0.0268</u>	0.0021	0.0330	0.0092	0.0655	0.0016	0.0293	<u>0.0015</u>	0.0265	0.0472	0.1762	0.0024	0.0342
	48	0.0024	0.0297	0.0205	0.0973	0.0034	0.0418	0.0030	<u>0.0336</u>	<u>0.0027</u>	0.0395	0.0472	0.1782	0.0028	0.0342
	96	0.0029	0.0350	0.0040	0.0403	0.0033	0.0420	0.0021	0.0296	0.0094	0.0808	0.0504	0.1841	<u>0.0027</u>	<u>0.0348</u>
Rideshare	24	<u>0.0131</u>	<u>0.0915</u>	0.0154	0.1068	0.2906	0.5208	2.3921	1.4642	0.0120	0.0888	0.0197	0.1111	0.0165	0.1099
	48	<u>0.0262</u>	<u>0.1213</u>	0.0416	0.1639	0.2737	0.4929	1.7329	1.1560	0.0410	0.1550	0.0150	0.0928	0.0277	0.1256
	96	<u>0.0319</u>	<u>0.1387</u>	0.1029	0.2372	0.9701	0.9255	2.3393	1.3725	0.0493	0.1750	0.0197	0.1074	0.1278	0.3022
Traffic-Hourly	24	<u>0.0018</u>	<u>0.0295</u>	0.0173	0.1024	0.0374	0.1318	0.0026	0.0436	0.0012	0.0251	0.0860	0.2614	0.0095	0.0675
	48	0.0149	0.0768	0.0173	0.0979	0.0273	0.1044	<u>0.0050</u>	<u>0.0561</u>	0.0041	0.0437	0.0616	0.2233	0.0118	0.0815
	96	0.0051	0.0434	0.1193	0.1853	0.0114	0.0738	<u>0.0053</u>	<u>0.0494</u>	0.0061	0.0542	0.0770	0.2451	0.0207	0.0818
Traffic-Weekly	24	1.8497	1.2024	2.4431	1.4154	1.1848	0.8930	3.0522	1.5480	<u>1.6256</u>	<u>1.1081</u>	3.4161	1.6974	2.0727	1.2638
	48	1.1519	0.8876	1.2136	0.8902	0.8192	0.7094	1.5264	1.0127	<u>0.9544</u>	<u>0.8198</u>	1.8517	1.1764	1.0226	0.8245
	96	<u>1.0823</u>	<u>0.8553</u>	1.8327	1.0821	2.3833	1.2494	1.9142	1.1064	1.0690	0.8526	2.5015	1.3948	2.2524	1.1786
Vehicle-Trips	24	0.0576	0.1869	0.4024	0.5118	0.1580	0.3008	<u>0.0631</u>	<u>0.2116</u>	0.0987	0.2677	0.9044	0.8252	0.8109	0.7874
	48	0.1789	0.2736	0.2840	0.3786	0.2412	0.3377	<u>0.2111</u>	0.3672	0.2204	0.3546	1.6465	1.1149	0.4339	0.5371
	96	2.1508	1.2384	0.6030	0.5931	0.2146	0.3184	0.6472	0.6222	<u>0.2683</u>	<u>0.4108</u>	1.4847	1.0545	1.6101	1.0623
1st Count		5	4	8	8	3	3	6	7	9	9	2	2	0	0

Position: Time-Series Foundation Models Require Explicit Domain-Level Benchmarks

Table 9. Retail Time Series Forecasting Results

Dataset	Horizon	Chronos		Sundial		TimeGPT		TimesFM		TimesMOE		ToTo		Moirai	
		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
Amazon Sales	24	1.1189	0.8237	1.3381	0.9377	1.1309	0.8490	1.1554	0.8185	1.1651	0.8796	1.1163	0.8896	1.1391	0.8611
	48	1.1114	0.8710	1.5703	1.0111	1.3044	0.9534	1.1346	<u>0.8714</u>	1.1361	0.8788	<u>1.1325</u>	0.9010	1.2480	0.9219
	96	1.0314	0.8127	1.1558	0.8933	1.0491	0.8158	1.0419	0.8166	1.0118	0.8076	1.1189	0.8772	1.0220	0.8022
Coffee Sales	24	1.1815	1.0154	0.5546	0.6779	0.8117	<u>0.5220</u>	2.9566	1.6405	<u>0.4987</u>	0.6553	1.2543	0.9358	0.4330	0.4738
	48	<u>0.5318</u>	0.6145	0.9103	0.8122	0.8826	<u>0.5358</u>	1.6943	1.0705	0.2430	0.4299	2.0315	1.1849	1.4152	1.0200
	96	2.1459	1.2822	2.9386	1.5485	2.2502	1.4357	2.2698	1.2863	0.8457	0.7865	<u>0.8880</u>	<u>0.8337</u>	1.6077	1.0214
Dominic Dataset	24	0.3175	0.2378	0.3718	0.3865	<u>0.3036</u>	0.3310	0.3411	0.3450	0.3485	0.3907	0.2369	<u>0.2453</u>	0.3625	0.3008
	48	0.3959	0.2994	0.4608	0.3218	0.2219	<u>0.3019</u>	0.4229	0.3490	0.3941	0.3360	<u>0.2901</u>	0.3910	0.4331	0.3100
	96	1.2030	0.4447	1.3115	0.6073	0.2020	0.1755	1.2137	0.4449	1.1841	0.4205	<u>0.2387</u>	<u>0.3214</u>	1.3643	0.5986
Online Retail	24	3.8387	1.2698	3.6936	1.1712	2.3687	<u>0.8337</u>	3.1464	1.0011	2.8458	0.9423	4.3766	1.2302	<u>2.7470</u>	0.7555
	48	2.7728	1.1570	2.6140	1.0291	<u>2.2017</u>	0.8566	2.8359	1.1480	2.1514	<u>0.9124</u>	5.6920	1.7018	2.5861	0.9965
	96	2.8534	1.2491	<u>2.1085</u>	<u>0.9790</u>	2.6106	1.1657	2.4595	1.1195	1.8616	0.9391	7.1069	1.9730	2.4705	1.1167
Product 1	24	0.5123	0.5252	0.3827	0.5188	<u>0.3156</u>	<u>0.4252</u>	0.2979	0.4205	0.4614	0.4909	0.7121	0.7402	0.4015	0.4943
	48	<u>0.3531</u>	<u>0.4375</u>	0.3651	0.4593	0.3941	0.4849	0.2794	0.4036	0.4824	0.5671	0.7598	0.7037	0.8853	0.7722
	96	0.3336	0.4130	0.4730	0.5288	0.4371	0.5101	<u>0.3601</u>	<u>0.4341</u>	0.4405	0.5217	0.7434	0.6279	0.6082	0.6306
Product 2	24	1.4701	0.8309	<u>0.9281</u>	0.6432	1.4052	0.7522	0.9330	0.5857	0.6630	<u>0.5953</u>	2.2956	1.2089	1.4474	0.7860
	48	0.6878	0.5032	<u>0.5240</u>	0.5486	0.7123	<u>0.4907</u>	0.5259	0.4301	0.4561	<u>0.5312</u>	1.9612	1.1200	0.6097	0.5293
	96	<u>0.4488</u>	0.4273	0.4766	0.5332	0.6029	0.6221	0.3904	<u>0.4336</u>	0.5427	0.6135	1.7259	1.0568	1.1421	0.9529
Product 3	24	0.2430	0.3841	0.2952	0.4077	0.2809	0.4150	0.2127	<u>0.3525</u>	<u>0.2284</u>	0.3482	0.3417	0.4694	0.2289	0.3753
	48	0.3433	0.4562	0.3125	0.4622	0.3356	0.4648	0.1991	0.3591	<u>0.2776</u>	<u>0.4397</u>	0.3570	0.5093	0.5111	0.6111
	96	0.2705	0.4199	0.3908	0.5004	0.6019	0.6537	<u>0.2825</u>	0.3931	0.3269	0.4600	0.4894	0.5950	0.8329	0.7570
Rossmann	24	0.0372	0.1454	0.1385	0.2774	0.3788	0.5036	<u>0.0995</u>	<u>0.2389</u>	0.1135	0.2598	0.8031	0.6247	0.6378	0.6172
	48	0.7374	0.5075	0.1598	0.2941	0.9394	0.7648	<u>0.1729</u>	<u>0.3386</u>	0.2341	0.3316	0.8451	0.6461	0.8942	0.6927
	96	1.5239	0.8828	0.4013	0.4440	1.7268	1.1196	<u>0.3531</u>	<u>0.3980</u>	0.4442	0.4925	0.8431	0.6614	1.0488	0.8528
Store 1 Item 1	24	0.9408	0.7918	<u>0.5298</u>	0.6322	0.7691	0.7205	0.4497	0.5114	0.5920	<u>0.5982</u>	0.6350	0.6114	1.1624	0.8264
	48	0.9529	0.8110	0.5294	<u>0.5703</u>	1.1941	0.9246	<u>0.5321</u>	0.5605	0.8406	0.7410	0.5917	0.6303	1.2059	0.8882
	96	1.1248	0.8710	<u>0.5549</u>	<u>0.5991</u>	1.0041	0.8123	0.4706	0.5587	0.7287	0.6823	1.0241	0.8162	1.2087	0.8937
Store 1 Item 2	24	0.3491	0.4473	<u>0.2112</u>	0.3578	0.5490	0.6615	0.1936	<u>0.3580</u>	0.4609	0.5672	0.4573	0.5458	0.9026	0.8769
	48	0.5814	0.6264	<u>0.3216</u>	<u>0.4057</u>	1.3197	1.0135	0.2505	0.3832	0.6347	0.6317	0.3491	0.4847	1.1870	0.9215
	96	0.4283	0.5424	<u>0.3854</u>	<u>0.4991</u>	0.9424	0.8093	0.2683	0.4110	0.5939	0.6147	0.6565	0.6481	1.0154	0.8249
Store 1 Item 3	24	0.7481	0.7503	0.2266	0.3793	0.5835	0.6408	<u>0.2275</u>	<u>0.4008</u>	0.2539	0.4154	0.6160	0.6964	0.8301	0.8015
	48	0.4022	0.5208	<u>0.3106</u>	<u>0.4519</u>	1.2101	0.8816	0.2105	0.3697	0.4287	0.5519	0.5365	0.6399	1.2804	0.9476
	96	0.7485	0.7046	<u>0.4019</u>	<u>0.5062</u>	0.7625	0.6910	0.3095	0.4347	0.4252	0.5415	0.8323	0.8076	1.1725	0.8842
Store 1 Item 4	24	0.4129	0.5265	<u>0.3829</u>	0.4811	0.4064	0.5255	0.2903	0.4178	0.5087	0.5886	0.3913	<u>0.4723</u>	0.8440	0.8006
	48	0.8730	0.7839	0.7090	0.6745	1.3243	0.9874	0.4155	0.5279	0.7267	0.7341	<u>0.6218</u>	<u>0.6651</u>	1.0521	0.8814
	96	0.9286	0.7925	<u>0.5806</u>	<u>0.6207</u>	0.7315	0.6841	0.5192	0.5887	0.6753	0.6590	0.6538	0.7096	0.9032	0.7868
Store 1 Item 5	24	<u>0.7915</u>	0.7175	0.8010	0.6973	1.0323	0.8681	0.5734	0.5838	0.9410	0.7890	0.7760	0.7325	1.2901	0.9378
	48	0.9105	0.7609	<u>0.7200</u>	<u>0.6586</u>	1.2275	0.9303	0.5916	0.6058	0.9369	0.7747	0.9014	0.7341	1.0309	0.7930
	96	0.8603	0.7385	<u>0.6601</u>	<u>0.6496</u>	1.7525	1.1024	0.5579	0.5804	0.8473	0.7306	1.0905	0.8385	1.0845	0.8424
Average		0.989	0.689	0.907	0.661	1.080	0.722	0.896	0.639	0.718	0.622	1.388	0.832	1.193	0.820
1st Count		4	6	4	4	3	2	18	20	7	4	2	0	1	3

B.2. Additional Results

Table 10. TabPFN-TS multidomain forecasting: Energy, Health, and Nature

ENERGY				HEALTH				NATURE			
Dataset	H	MSE	MAE	Dataset	H	MSE	MAE	Dataset	H	MSE	MAE
AUS_Elec_Demand	24	0.0437	0.1796	COVID-ICU	24	0.00014	0.0072	Indian-Climate-AQI	24	0.9824	0.8917
AUS_Elec_Demand	48	0.0338	0.1519	COVID-ICU	48	0.00063	0.0221	Indian-Climate-AQI	48	0.9735	0.8721
AUS_Elec_Demand	96	0.0246	0.1287	COVID-ICU	96	0.00118	0.0292	Indian-Climate-AQI	96	0.9587	0.8412
Electricity_Weekly	24	0.5413	0.3332	COVID-Deaths	24	0.0019	0.0368	Indian-Climate-Humidity	24	0.9186	0.7963
Electricity_Weekly	48	0.5096	0.3294	COVID-Deaths	48	0.0587	0.2183	Indian-Climate-Humidity	48	0.8124	0.7510
Electricity_Weekly	96	0.4878	0.3321	COVID-Deaths	96	0.3742	0.4765	Indian-Climate-Humidity	96	0.8697	0.8026
ETTh1	24	0.1669	0.3287	COVID-Weekly	24	0.00013	0.0056	Indian-Climate-Temperature	24	0.9483	0.8724
ETTh1	48	0.2793	0.4182	COVID-Weekly	48	0.00027	0.0045	Indian-Climate-Temperature	48	0.9621	0.8459
ETTh1	96	0.2226	0.3578	COVID-Weekly	96	0.00042	0.0161	Indian-Climate-Temperature	96	0.9385	0.8328
ETTh2	24	0.1813	0.3761	Hospital	24	0.3627	0.5021	Indian-Climate-Wind	24	1.1216	0.9215
ETTh2	48	0.1514	0.3311	Hospital	48	0.6384	0.6273	Indian-Climate-Wind	48	1.0428	0.8736
ETTh2	96	0.2072	0.3439	Hospital	96	0.5286	0.5979	Indian-Climate-Wind	96	1.0127	0.8713
ETTm1	24	0.2187	0.3526	Chickenpox	24	0.2893	0.2584	Jena-Climate-H2OC	24	0.1243	0.3015
ETTm1	48	0.3018	0.4417	Chickenpox	48	0.6671	0.4472	Jena-Climate-H2OC	48	0.0528	0.1948
ETTm1	96	0.1342	0.2538	Chickenpox	96	0.5476	0.5179	Jena-Climate-H2OC	96	0.2516	0.4683
ETTm2	24	0.0739	0.2296	Illness	24	0.2718	0.4376	Jena-Climate-RH	24	0.0016	0.0352
ETTm2	48	0.2964	0.4912	Illness	48	0.1036	0.2681	Jena-Climate-RH	48	0.0058	0.0651
ETTm2	96	0.2214	0.3967	Illness	96	0.0974	0.2588	Jena-Climate-RH	96	0.0119	0.0863
London_SmartMeters	24	1.9348	0.9312	T1	24	1.9734	1.2862	Ozone	24	0.2037	0.3495
London_SmartMeters	48	1.7683	0.8786	T1	48	2.3417	1.2289	Ozone	48	0.1854	0.3728
London_SmartMeters	96	1.0715	0.5984	T1	96	0.9146	0.8863	Ozone	96	0.2986	0.4481
Solar_10min	24	0.0013	0.0236	T2	24	0.0317	0.1234	Pollution	24	0.0065	0.0614
Solar_10min	48	0.0046	0.0352	T2	48	0.0826	0.2078	Pollution	48	0.0789	0.2086
Solar_10min	96	0.1328	0.3224	T2	96	0.0982	0.2731	Pollution	96	0.3487	0.4622
				T3	24	0.1986	0.3682	Saugeen-Day	24	1.3428	0.6127
				T3	48	0.2874	0.4475	Saugeen-Day	48	1.0894	0.6025
				T3	96	0.4168	0.5186	Saugeen-Day	96	0.6083	0.3986
				T4	24	1.4879	0.6974	Sunspot	24	0.00012	0.0031
				T4	48	0.8362	0.6183	Sunspot	48	0.0036	0.0218
				T4	96	0.8325	0.6651	Sunspot	96	0.0049	0.0267
				US-Births	24	0.1974	0.3086	Temp-Hourly	24	0.0269	0.1385
				US-Births	48	0.1769	0.2728	Temp-Hourly	48	0.1042	0.2683
				US-Births	96	0.1528	0.2136	Temp-Hourly	96	0.2215	0.4028
								Weather	24	0.0349	0.0826
								Weather	48	0.0362	0.0821
								Weather	96	0.0403	0.0835

Table 11. TabPFN-TS multidomain forecasting: Retail, Transport, and Finance

RETAIL				TRANSPORT				FINANCE			
Dataset	H	MSE	MAE	Dataset	H	MSE	MAE	Dataset	H	MSE	MAE
Amazon Sales	24	1.0954	0.8126	Cross-1	24	0.0287	0.1489	AAL	24	0.0012	0.0271
Amazon Sales	48	1.1548	0.8912	Cross-1	48	0.0345	0.1598	AAL	48	0.0019	0.0324
Amazon Sales	96	1.0473	0.8295	Cross-1	96	0.0589	0.1867	AAL	96	0.0026	0.0398
Coffee Sales	24	0.4479	0.4926	Cross-2	24	0.0669	0.2035	AAPL	24	0.0258	0.1423
Coffee Sales	48	0.2845	0.4688	Cross-2	48	0.0478	0.1692	AAPL	48	0.0267	0.1268
Coffee Sales	96	0.9013	0.8284	Cross-2	96	0.1184	0.2661	AAPL	96	0.0459	0.1682
Dominic Dataset	24	0.2386	0.2498	Cross-3	24	0.0396	0.1688	BLDP	24	0.0013	0.0325
Dominic Dataset	48	0.3291	0.3365	Cross-3	48	0.0637	0.2049	BLDP	48	0.0029	0.0481
Dominic Dataset	96	0.2214	0.2336	Cross-3	96	0.1042	0.2684	BLDP	96	0.0016	0.0298
Online Retail	24	2.7486	0.8623	Cross-4	24	0.0738	0.2136	CASH	24	0.0112	0.0887
Online Retail	48	2.3517	0.9441	Cross-4	48	0.1012	0.2587	CASH	48	0.1586	0.3724
Online Retail	96	2.0863	0.9894	Cross-4	96	0.1276	0.2825	CASH	96	0.0219	0.1183
Product 1	24	0.3298	0.4442	Cross-5	24	0.0648	0.1896	CBD	24	0.0010	0.0258
Product 1	48	0.3015	0.4287	Cross-5	48	0.0961	0.2442	CBD	48	0.0015	0.0296
Product 1	96	0.3724	0.4521	Cross-5	96	0.0897	0.2334	CBD	96	0.0038	0.0527
Product 2	24	0.6847	0.5983	Cross-6	24	0.0472	0.1621	CIF2016	24	0.1685	0.3521
Product 2	48	0.4722	0.4398	Cross-6	48	0.0483	0.1702	CIF2016	48	0.2627	0.4018
Product 2	96	0.3965	0.4311	Cross-6	96	0.0712	0.2179	CIF2016	96	0.9124	0.7816
Product 3	24	0.2358	0.3725	Pedestrian-Counts	24	0.0014	0.0261	DLR	24	0.0392	0.1786
Product 3	48	0.2624	0.4086	Pedestrian-Counts	48	0.0041	0.0368	DLR	48	0.0195	0.1127
Product 3	96	0.3087	0.4193	Pedestrian-Counts	96	0.0033	0.0362	DLR	96	0.1817	0.3524
Rossmann	24	0.0496	0.1658	Rideshare	24	0.0164	0.1018	EDUC	24	0.0084	0.0759
Rossmann	48	0.2217	0.3369	Rideshare	48	0.0172	0.0987	EDUC	48	0.0136	0.0948
Rossmann	96	0.4583	0.4726	Rideshare	96	0.0239	0.1164	EDUC	96	0.0171	0.1032
Store 1 Item 1	24	0.4825	0.5489	Traffic-Hourly	24	0.0010	0.0238	GWG	24	0.0108	0.0895
Store 1 Item 1	48	0.5736	0.5897	Traffic-Hourly	48	0.0046	0.0448	GWG	48	0.1246	0.3142
Store 1 Item 1	96	0.5124	0.5882	Traffic-Hourly	96	0.0057	0.0512	GWG	96	0.4987	0.6289
Store 1 Item 2	24	0.2243	0.3784	Traffic-Weekly	24	1.8124	1.1843	HAS	24	0.0116	0.0869
Store 1 Item 2	48	0.2915	0.4178	Traffic-Weekly	48	0.8876	0.7489	HAS	48	0.1042	0.2813
Store 1 Item 2	96	0.3097	0.4362	Traffic-Weekly	96	1.1987	0.8615	HAS	96	0.2489	0.4017
Store 1 Item 3	24	0.2486	0.4087	Vehicle-Trips	24	0.0587	0.1981	NN5-Daily	24	0.2129	0.4058
Store 1 Item 3	48	0.2379	0.3948	Vehicle-Trips	48	0.2074	0.3076	NN5-Daily	48	0.2413	0.4117
Store 1 Item 3	96	0.3412	0.4596	Vehicle-Trips	96	0.2638	0.3449	NN5-Daily	96	0.3526	0.4712
Store 1 Item 4	24	0.3297	0.4485					NN5-Weekly	24	1.0845	0.9036
Store 1 Item 4	48	0.4682	0.5614					NN5-Weekly	48	1.6892	1.1984
Store 1 Item 4	96	0.5716	0.6142					NN5-Weekly	96	0.5123	0.6648
Store 1 Item 5	24	0.6428	0.6219					NSC	24	0.0265	0.1482
Store 1 Item 5	48	0.6735	0.6412					NSC	48	0.0412	0.1689
Store 1 Item 5	96	0.6187	0.6276					NSC	96	0.0328	0.1467
								PHG	24	0.0156	0.1093
								PHG	48	0.0437	0.1964
								PHG	96	0.0918	0.2597

Table 12. MOMENT multidomain forecasting: Energy, Finance, and Healthcare

ENERGY				FINANCE				HEALTHCARE			
Dataset	H	MSE	MAE	Dataset	H	MSE	MAE	Dataset	H	MSE	MAE
AUS_Elec_Demand	24	0.2752	0.4476	AAL	24	0.0639	0.2494	COVID_ICU_Hospitalizations	24	0.0341	0.1822
AUS_Elec_Demand	48	0.6768	0.6474	AAL	48	0.0776	0.2714	COVID_ICU_Hospitalizations	48	0.0274	0.1592
AUS_Elec_Demand	96	0.6921	0.6609	AAL	96	0.0645	0.2446	COVID_ICU_Hospitalizations	96	0.018	0.1106
Electricity_Weekly	24	1.4592	1.0324	AAPL	24	0.0242	0.1317	COVID_Total_Cases	24	0.0	0.0031
Electricity_Weekly	48	0.9863	0.8604	AAPL	48	0.0237	0.1328	COVID_Total_Cases	48	0.0	0.0059
Electricity_Weekly	96	1.1454	0.8461	AAPL	96	0.2345	0.3967	COVID_Total_Cases	96	0.0003	0.0163
ETTh1	24	2.0849	1.3048	BLDP	24	0.2644	0.5081	COVID_Deaths	24	4.8141	2.1916
ETTh1	48	1.8796	1.2095	BLDP	48	0.2938	0.5378	COVID_Deaths	48	4.8532	2.1789
ETTh1	96	1.622	1.0842	BLDP	96	0.2575	0.5007	COVID_Deaths	96	3.3692	1.6152
ETTh2	24	0.6918	0.7373	CASH	24	0.0439	0.1781	COVID_Weekly_Deaths	24	0.0185	0.1334
ETTh2	48	0.4787	0.5855	CASH	48	0.2102	0.3659	COVID_Weekly_Deaths	48	0.0285	0.1662
ETTh2	96	0.3059	0.4354	CASH	96	0.3689	0.5139	COVID_Weekly_Deaths	96	0.0507	0.2227
ETTm1	24	0.9809	0.8259	CBD	24	0.0594	0.2387	Hospital	24	0.5198	0.5882
ETTm1	48	2.6684	1.4375	CBD	48	0.0719	0.2628	Hospital	48	0.828	0.7492
ETTm1	96	1.9763	1.2647	CBD	96	0.117	0.3347	Hospital	96	–	–
ETTm2	24	0.6748	0.7947	CIF2016	24	3.0636	1.731	Hungary_Chickenpox	24	1.2567	1.0477
ETTm2	48	0.904	0.8966	CIF2016	48	3.0205	1.6775	Hungary_Chickenpox	48	1.1316	0.8492
ETTm2	96	0.7852	0.7791	CIF2016	96	3.5369	1.7017	Hungary_Chickenpox	96	0.9479	0.8053
London_SmartMeters	24	3.7717	1.4318	DLR	24	0.443	0.6581	National_Illness	24	0.4093	0.5633
London_SmartMeters	48	3.0155	1.2884	DLR	48	0.7587	0.8517	National_Illness	48	0.4931	0.6341
London_SmartMeters	96	1.7771	0.9638	DLR	96	0.5797	0.6983	National_Illness	96	1.7433	0.9834
Solar_10min	24	0.2422	0.4794	EDUC	24	4.4523	2.0944	T1	24	3.1012	1.6472
Solar_10min	48	0.229	0.4635	EDUC	48	6.4124	2.51	T1	48	2.2389	1.3064
Solar_10min	96	0.2202	0.4255	EDUC	96	8.1416	2.8409	T1	96	2.9999	1.5613
				GWW	24	1.3954	1.1771	T2	24	0.0901	0.2541
				GWW	48	1.1287	1.0054	T2	48	0.5876	0.5735
				GWW	96	1.2641	1.0821	T2	96	1.3177	0.8817
				HAS	24	0.9749	0.9835	T3	24	1.183	0.9925
				HAS	48	0.8802	0.9298	T3	48	1.2475	1.0514
				HAS	96	0.5387	0.6768	T3	96	0.8653	0.8116
				NN5_Daily	24	1.7809	0.9837	T4	24	1.6642	0.9313
				NN5_Daily	48	1.5961	0.9411	T4	48	1.1221	0.8307
				NN5_Daily	96	1.6968	1.0002	T4	96	0.891	0.7734
				NN5_Weekly	24	3.1347	1.6668	US_Births	24	1.6167	1.0388
				NN5_Weekly	48	3.0486	1.6218	US_Births	48	1.4243	0.9312
				NN5_Weekly	96	2.594	1.3651	US_Births	96	1.3049	0.9738
				NSC	24	0.0225	0.1175				
				NSC	48	0.1509	0.3168				
				NSC	96	0.0994	0.234				
				PHG	24	3.4251	1.8452				
				PHG	48	4.0593	2.0066				
				PHG	96	4.377	2.0773				

Table 13. MOMENT multidomain forecasting: Nature, Retail, and Transport

NATURE				RETAIL				TRANSPORT			
Dataset	H	MSE	MAE	Dataset	H	MSE	MAE	Dataset	H	MSE	MAE
Indian_Climate_AQI	24	1.138	0.9562	Amazon_Sales	24	1.2132	0.8843	Cross_1	24	0.3387	0.5051
Indian_Climate_AQI	48	1.0073	0.9008	Amazon_Sales	48	1.1731	0.904	Cross_1	48	0.2793	0.456
Indian_Climate_AQI	96	1.0451	0.8914	Amazon_Sales	96	1.0382	0.7975	Cross_1	96	0.7275	0.7296
Indian_Climate_Humidity	24	1.0232	0.8742	Coffee_Sales	24	3.4196	1.804	Cross_2	24	0.6832	0.7568
Indian_Climate_Humidity	48	1.1192	0.9234	Coffee_Sales	48	3.347	1.7754	Cross_2	48	0.4517	0.538
Indian_Climate_Humidity	96	1.1471	0.9181	Coffee_Sales	96	3.156	1.6173	Cross_2	96	0.9623	0.8086
Indian_Climate_Temperature	24	1.0286	0.918	Dominic_Dataset	24	0.3751	0.4401	Cross_3	24	0.2549	0.3828
Indian_Climate_Temperature	48	1.0614	0.8827	Dominic_Dataset	48	0.4657	0.3777	Cross_3	48	0.2026	0.3503
Indian_Climate_Temperature	96	1.0237	0.8739	Dominic_Dataset	96	1.0606	0.4575	Cross_3	96	0.5052	0.5584
Indian_Climate_Wind	24	1.2106	0.9383	Online_Retail	24	5.1241	1.6251	Cross_4	24	0.3183	0.4666
Indian_Climate_Wind	48	1.3136	0.9883	Online_Retail	48	3.2703	1.2814	Cross_4	48	0.3274	0.4901
Indian_Climate_Wind	96	1.1546	0.9375	Online_Retail	96	2.0988	0.9942	Cross_4	96	0.9552	0.8039
Jena_Climate_H2OC	24	0.7598	0.8432	Product_1	24	0.6292	0.6473	Cross_5	24	0.1685	0.3206
Jena_Climate_H2OC	48	1.1281	1.0282	Product_1	48	0.6915	0.6801	Cross_5	48	0.3388	0.4645
Jena_Climate_H2OC	96	1.43	1.0651	Product_1	96	0.7058	0.6807	Cross_5	96	0.7979	0.6908
Jena_Climate_RH	24	0.259	0.5041	Product_2	24	1.3392	0.9013	Cross_6	24	0.2433	0.381
Jena_Climate_RH	48	0.2213	0.4602	Product_2	48	1.0277	0.8445	Cross_6	48	0.1441	0.2869
Jena_Climate_RH	96	0.2801	0.5126	Product_2	96	0.8973	0.7799	Cross_6	96	0.283	0.4436
Ozone	24	0.1639	0.3244	Product_3	24	0.9471	0.8559	Pedestrian_Counts	24	0.0271	0.142
Ozone	48	0.3733	0.5055	Product_3	48	0.9303	0.8403	Pedestrian_Counts	48	0.0213	0.126
Ozone	96	0.2742	0.4232	Product_3	96	0.8487	0.7755	Pedestrian_Counts	96	0.0278	0.1424
Pollution	24	0.9063	0.9273	Rossmann	24	0.8884	0.741	Rideshare	24	0.7778	0.869
Pollution	48	0.6886	0.7472	Rossmann	48	1.0714	0.7926	Rideshare	48	0.8692	0.9048
Pollution	96	1.7419	0.9876	Rossmann	96	1.2495	0.9068	Rideshare	96	0.9849	0.9672
Saugeen_Day	24	1.5539	0.7213	Store_1_Item_1	24	1.2051	0.9026	Traffic_Hourly	24	0.1943	0.3716
Saugeen_Day	48	1.1927	0.6815	Store_1_Item_1	48	1.2329	0.9632	Traffic_Hourly	48	0.2139	0.3713
Saugeen_Day	96	0.625	0.4695	Store_1_Item_1	96	0.9288	0.7909	Traffic_Hourly	96	0.2495	0.401
Sunspot	24	0.0021	0.044	Store_1_Item_2	24	1.0651	0.9435	Traffic_Weekly	24	2.1159	1.2938
Sunspot	48	0.0071	0.0584	Store_1_Item_2	48	0.9592	0.8504	Traffic_Weekly	48	1.2372	0.9211
Sunspot	96	0.006	0.0619	Store_1_Item_2	96	0.7729	0.7253	Traffic_Weekly	96	1.7488	1.0553
Temp_Hourly	24	1.2862	0.9748	Store_1_Item_3	24	1.356	0.9918	Vehicle_Trips	24	1.1097	0.9459
Temp_Hourly	48	1.2445	0.9948	Store_1_Item_3	48	1.049	0.825	Vehicle_Trips	48	1.1148	0.9414
Temp_Hourly	96	1.141	0.9428	Store_1_Item_3	96	0.7756	0.696	Vehicle_Trips	96	1.051	0.927
Weather	24	0.1213	0.2762	Store_1_Item_4	24	1.2675	1.0148				
Weather	48	0.0896	0.235	Store_1_Item_4	48	1.0296	0.8873				
Weather	96	0.1106	0.2668	Store_1_Item_4	96	0.8538	0.7514				
				Store_1_Item_5	24	1.5992	1.0624				
				Store_1_Item_5	48	1.2378	0.9033				
				Store_1_Item_5	96	1.0093	0.7937				

B.3. Aggregated Performance and Ranking

Here we report the same ranking analyses as in Section 4.2, but using the MSE (Mean Squared Error) metric. Figure 5 shows model rankings by domain (sorted by global mean performance), and Figure 6 presents the domain-specific performance footprint under MSE. The findings mirror those observed with MAE, with only minor deviations.

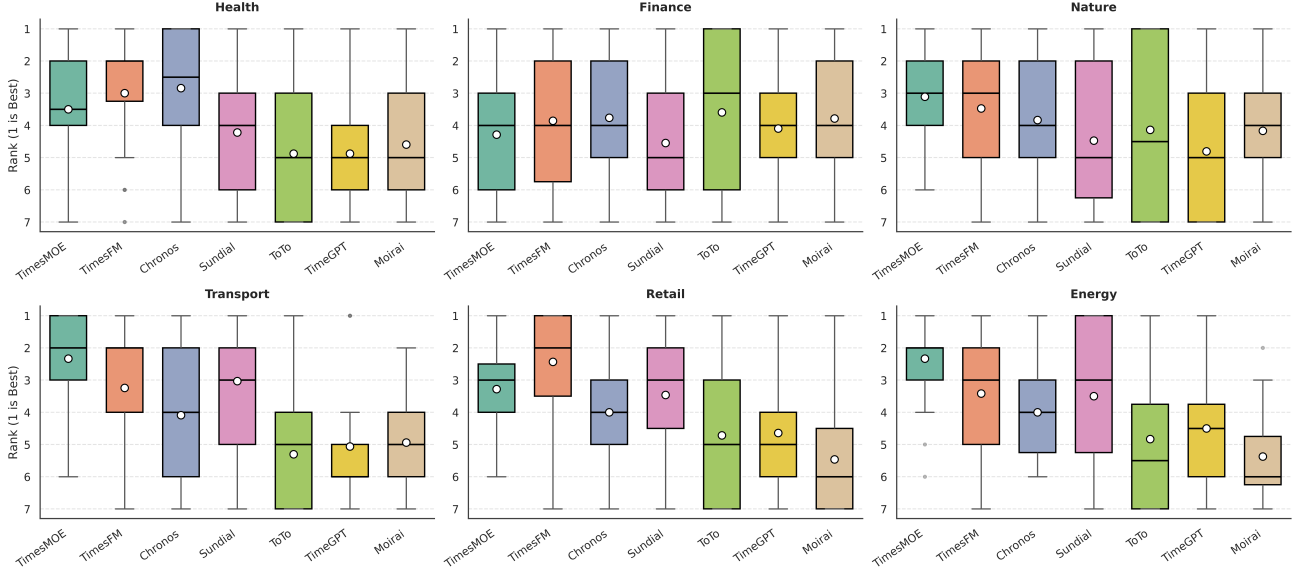


Figure 5. Model Rank Distribution (MSE) Across Domains. Box plots show the distribution of model rankings within each domain, where rank 1 indicates best performance. White circles denote mean rank; horizontal lines indicate median rank.

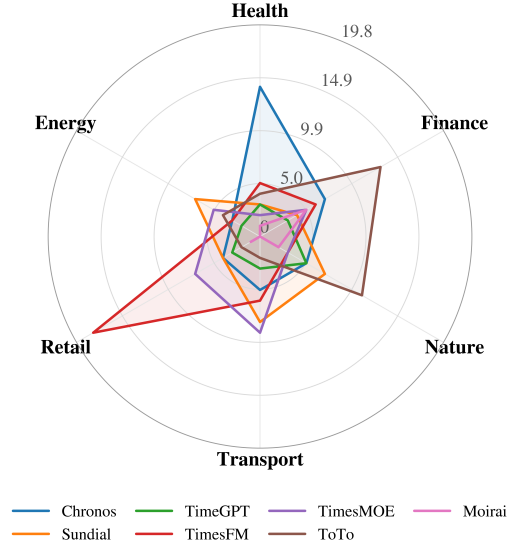


Figure 6. Domain-Specific Performance Footprint (MSE) of TSFMs

C. Statistical Test for Domain-Specific Performance Differences

C.1. Mathematical Formulation

Let $\mathcal{D} = \{d_1, \dots, d_K\}$ denote the set of domains, and let $\mathcal{M}$ denote the set of models. For a fixed forecasting horizon $h = 24$ and error metric $E \in \{\text{MAE}, \text{MSE}\}$, we collect per-dataset errors produced by model $m \in \mathcal{M}$ within each domain $d \in \mathcal{D}$. Concretely, if domain d contains n_d datasets (or benchmark instances), the observed error samples are

$$\mathbf{x}_d^{(m,E)} = \{x_{d,1}^{(m,E)}, x_{d,2}^{(m,E)}, \dots, x_{d,n_d}^{(m,E)}\}. \quad (7)$$

We test whether the distribution of errors is invariant across domains:

$$H_0 : \mathbf{x}_{d_1}^{(m,E)} \stackrel{d}{=} \mathbf{x}_{d_2}^{(m,E)} \stackrel{d}{=} \dots \stackrel{d}{=} \mathbf{x}_{d_K}^{(m,E)}, \quad H_1 : \exists d_i \neq d_j \text{ s.t. } \mathbf{x}_{d_i}^{(m,E)} \not\stackrel{d}{=} \mathbf{x}_{d_j}^{(m,E)}. \quad (8)$$

Since error distributions can be non-Gaussian and heteroskedastic across domains, we adopt the non-parametric Kruskal–Wallis test (Kruskal & Wallis, 1952). Let $N = \sum_{d \in \mathcal{D}} n_d$ be the total number of observations, and let $R(x)$ be the rank of observation x among all pooled samples $\cup_d \mathbf{x}_d^{(m,E)}$ (ties are assigned average ranks). Define the mean rank of each domain as

$$\bar{R}_d = \frac{1}{n_d} \sum_{i=1}^{n_d} R(x_{d,i}^{(m,E)}). \quad (9)$$

The Kruskal–Wallis statistic is then computed as

$$H = \frac{12}{N(N+1)} \sum_{d \in \mathcal{D}} n_d \bar{R}_d^2 - 3(N+1), \quad (10)$$

which is approximately χ^2 -distributed with $K - 1$ degrees of freedom under H_0 . We report the corresponding p -value for each model and metric, and annotate significance using the convention:

$$\text{ns} : p > 0.05, \quad * : p \leq 0.05, \quad ** : p \leq 0.025, \quad *** : p \leq 0.01, \quad **** : p \leq 10^{-3}. \quad (11)$$

C.2. Domain-Specific Performance Differences Across Horizons

C.2.1. HORIZON = 48

Table 14 demonstrates that domain-dependent performance differences remain statistically detectable at horizon $h = 48$ for most time-series foundation models. Under MAE, five out of seven models show significant cross-domain variation, with TIMESMOE exhibiting the strongest evidence ($p \leq 0.01$), while TIMESFM and ToTo do not reach significance. Under MSE, the domain effect is more consistent and stronger: all models achieve significance, and multiple models reach $p \leq 0.01$ (notably TIMEGPT and TIMESMOE), indicating that cross-domain shifts increasingly manifest as larger-magnitude errors at longer horizons. Overall, the results reinforce that aggregate benchmark reporting can mask domain-specific failure modes, motivating domain-aware evaluation when comparing TSFMs in realistic deployment settings.

Table 14. Global domain effect at Horizon = 48 using the Kruskal–Wallis test. Reported values are p -values with significance markers: ns ($p > 0.05$), * ($p \leq 0.05$), ** ($p \leq 0.025$), *** ($p \leq 0.01$).

Model	MAE (p , sig.)	MSE (p , sig.)
Chronos	0.0402*	0.0131**
Moirai	0.0492*	0.0106**
Sundial	0.0216**	0.0125**
TimeGPT	0.0294*	0.0064***
TimesFM	0.0806 (ns)	0.0282*
TimesMOE	0.0056***	0.0047***
ToTo	0.0528 (ns)	0.0248**

C.2.2. HORIZON = 96

Table 15 shows that domain-dependent performance differences become less uniformly detectable at the long-horizon setting ($h = 96$). Under MAE, only MOIRAI and TIMESMOE exhibit statistically significant cross-domain variation, whereas CHRONOS, SUNDIAL, TIMEGPT, TIMESFM, and TOTOT are not significant. Under MSE, evidence of domain dependence persists for SUNDIAL, TIMEGPT, MOIRAI, and TIMESMOE, while CHRONOS, TIMESFM, and TOTOT remain non-significant. Overall, the weakening of significance at $h = 96$ suggests that long-range uncertainty may dominate forecasting errors for some models, effectively compressing domain-specific separability; nevertheless, the consistent significance of TIMESMOE indicates that certain architectures still retain strong domain sensitivity even at extended horizons.

Table 15. Global domain effect at Horizon = 96 using the Kruskal–Wallis test. Reported values are p -values with significance markers: *ns* ($p > 0.05$), * ($p \leq 0.05$), ** ($p \leq 0.025$), *** ($p \leq 0.01$).

Model	MAE (p , sig.)	MSE (p , sig.)
Chronos	0.2064 (<i>ns</i>)	0.0668 (<i>ns</i>)
Moirai	0.0242**	0.0184**
Sundial	0.0544 (<i>ns</i>)	0.0293*
TimeGPT	0.0518 (<i>ns</i>)	0.0213**
TimesFM	0.5652 (<i>ns</i>)	0.4340 (<i>ns</i>)
TimesMOE	0.0077***	0.0077***
ToTo	0.4542 (<i>ns</i>)	0.2521 (<i>ns</i>)

C.2.3. TRENDS IN DOMAIN DEPENDENCE IN DIFFERENT HORIZONS

Across horizons, the Kruskal–Wallis tests reveal a clear attenuation of domain-dependent performance differences as the forecast horizon increases from $h = 24$ to $h = 96$.

At short horizon ($h = 24$), domain effects are pervasive: nearly all models reject the null hypothesis under MSE, and most remain significant under MAE, indicating that cross-domain statistical heterogeneity (e.g., noise structure, seasonality strength, and missingness patterns) translates into measurable differences in error distributions. At mid-range horizon ($h = 48$), the signal remains strong under MSE (all models significant), while MAE begins to weaken for some methods (e.g., TIMESFM, TOTOT), suggesting that domain shifts increasingly manifest through occasional large deviations rather than systematic median-level shifts.

By long horizon ($h = 96$), significance becomes sparse and model-dependent: several methods become non-significant for both metrics, consistent with the hypothesis that long-range uncertainty and compounding multi-step errors increasingly dominate the loss, inflating error variance and causing domain-specific effects to be masked by a higher noise floor. Consequently, rank separation across domains becomes weaker at larger horizons, yielding systematically reduced statistical confidence (larger p -values) as h increases.

Overall, the horizon-wise trend indicates that domain-aware evaluation is most critical at short-to-mid horizons where cross-domain differences are strongest, while at extended horizons only a subset of models continues to exhibit statistically distinguishable domain-specific performance.